WHICH BENCHMARKS TO RUN? SELECTING AND INFERRING BENCHMARK SCORES FOR A NEW MODEL UNDER A BUDGET OF k BENCHMARKS

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ABSTRACT

Evaluating a new model on every benchmark of a leaderboard is costly, but historical score matrices of earlier models are available. We study the cold-start question: given such a model $\times$ benchmark matrix with missing cells and a new model that can run only k benchmarks, which k should it run, how should the other scores be inferred, and how large must k be? We evaluate on held-out models, with selection and all fitting restricted to training models, a fixed target set so that every k is scored on the same cells, bootstrap intervals over held-out models (checked against organisation- and family-clustered resampling), and a criterion for "enough" fixed in advance (the fraction of the gap between the column mean and a full-information predictor that is closed). The quantitative answers come mainly from embedding models (MTEB, nearly complete blocks of up to 275 tasks) under random 10-fold splits; five generative-LLM matrices ($N = 6$–57) give short curves. On a sparse MTEB matrix (30% observed) about 16 benchmarks are needed and informed selection no longer beats random ones from $k = 16$. On MTEB (152 models $\times$ 56 tasks, random folds) four informed benchmarks give a normalised error of 0.20–0.22 against 0.30 for random ones and 1.04 for the column mean, and greedy selection meets the gap-closed criterion at $k = 2$–4 (for 80%) and $k = 4$–16 (for 90%) over two selectors and two inference rules, on a coarse grid of k (random selection: 5–8 and 12–16). Benchmarks chosen by conditional variance or log-determinant beat random ones at small k, whereas leverage-score, pivoted-QR and highest-variance selection are worse than random; the advantage shrinks to ≤ 0.03 nMSE by $k \approx 16$ and no selector is uniformly best (forward ridge and an adaptive greedy rule gain a further 0.03–0.04). With whole organisations held out the floors rise (from 0.15–0.18 to 0.27–0.33), the advantage of informed selection is gone for $k \geq 16$, and which informed selector needs fewest benchmarks changes with the split; inference fails for the newest models in a temporal split, where release-date terms do not help, but a Mahalanobis check on the revealed scores flags the MTEB-56 models with 3–11$\times$ the error. Bounds for random subsets and a submodularity guarantee for greedy selection describe the shape of the curves, not their level on real leaderboards. All numbers come from the shipped code.

1 INTRODUCTION

Leaderboards report each model on dozens to hundreds of benchmarks, and running a new model on all of them is expensive. Yet the scores of earlier models are already known, and benchmark scores of different models are strongly correlated: the score matrices studied here have a leading component that carries 36–80% of the variance. A natural practice is therefore to run a few benchmarks and infer the rest. Work on cheaper evaluation has mostly reduced the number of *items* inside one benchmark (Polo et al., 2024b; Kipnis et al., 2024), or fitted low-dimensional capability models to scores of many models (Ruan et al., 2024; Polo et al., 2024a). We ask the complementary question at the level of whole benchmarks, in the setting an evaluator faces for a model that has nothing in the table yet:

Given a historical score matrix with missing cells, and a new model that can run only k benchmarks: which k should it run, how should the remaining scores be inferred, and how large must k be?

The three parts interact, and "how large" has no meaning without a yardstick for "enough". We therefore fix the evaluation before looking at final results: the unit of evaluation is a held-out model; the selection order and every fitted quantity use training models only; the targets are a fixed set of benchmarks that no selector may pick, so every k is judged on the same cells; uncertainty is a bootstrap over held-out models; and k is called enough when it closes 80% or 90% of the gap between the column mean and the full-information predictor of the same inference rule (lower confidence bound), with absolute tolerances as a secondary criterion.

Contributions.

- A cold-start protocol and a pre-fixed criterion for the size of k (Sections 3, 4), applied to nine real matrices (four MTEB blocks of embedding models, one of them 30% observed; generative LLMs on HELM-MMLU, two HELM Lite matrices, LiveBench and the Open LLM Leaderboard) and to synthetic matrices with known truth.
- An empirical comparison of 10 selectors and 7 inference rules. Informed selection that accounts for redundancy (greedy conditional variance, greedy log-determinant) beats random selection by 0.08–0.15 nMSE at $k \leq 4$ on MTEB-56; common column-subset heuristics (leverage scores, pivoted QR, highest variance) do not, and a one-line centrality heuristic matches them only at $k \leq 2$. Direct contrasts show that no informed selector is uniformly best and the advantage is ≤ 0.03 nMSE by $k \approx 16$ (Section 5).
- Answers to "how large k": a handful closes most of the gap ($k = 2$–4 for 80% and $k = 4$–16 for 90% on MTEB-56, informed selection) to the full-information predictor, but that predictor is itself far from perfect ($\mathrm{nMSE} = 0.15$–0.36), and approaching it takes tens of benchmarks; the bootstrap interval of $k^\star$ and its value under leave-organisation-out and leave-family-out splits are reported, and the ranking of informed selectors is not stable across them.
- Stress tests (missing cells, few training models, rank, leave-family-out and temporal shift, leakage, noise, cost), with failures reported (Sections 5, 6).
- Bounds for random subsets and a $(1 - 1/e)$ guarantee for greedy information selection, checked against synthetic and real matrices (Appendix A, Fig. 8). They are a sanity check on the form of the k-curve, not a contribution about its level on real data.

All numbers come from the code shipped with the paper (Appendix C).

2 Related work

Cheaper evaluation inside a benchmark. tinyBenchmarks (Polo et al., 2024b) and metabench (Kipnis et al., 2024) shrink the number of *items* used to estimate a model's score on a given benchmark; metabench selects items with item-response-theory (IRT) information, reconstructs the scores of six benchmarks from a sparse item set, and compares against random subsets of equal size. We borrow the random-subset null and the "reconstruct what was not run" framing, but our unit of selection is the whole benchmark: the evaluator picks which benchmarks to run, and the remaining benchmarks are *not* run at all. Item-level reduction and benchmark-level reduction are complementary.

Low-dimensional structure of model capabilities. Observational scaling laws (Ruan et al., 2024) extract a few principal "capability" dimensions from a benchmark×model table and use them to predict downstream metrics; Sloth (Polo et al., 2024a) fits low-dimensional skill models with family-specific scaling and needs model-side covariates (size, tokens) and at least one model of the target family. We use the same structural idea (a low-rank latent space) but in a cold-start setting: the covariates of the new model are its *measured* scores on k benchmarks, no model metadata is used, and the question is which k benchmarks to measure and how many. Perlitz et al. (2024) show that benchmark agreement conclusions depend on methodological choices; our protocol therefore states its choices (splits, targets, normalisation) and varies them.

Subset selection and optimal design. Choosing columns of a matrix that best represent the rest is the column subset selection / CUR problem (Mahoney & Drineas, 2009), with leverage-score and pivoted-QR heuristics. Under a Gaussian model, choosing observations that minimise the posterior variance of unobserved quantities is a classical sensor-placement problem; greedy mutual-information or log-determinant selection has a $(1-1/e)$ guarantee for monotone submodular objectives (Nemhauser et al., 1978; Guestrin et al., 2005), and spectral conditions govern greedy variance reduction (Das & Kempe, 2011). Randomly pivoted Cholesky (Chen et al., 2025) is a recent relative. We use these as selectors and check, rather than assume, that they help on real score matrices.

Missing data and cold start. Our inference rules build on factor analysis and PCA with missing values (Ilin & Raiko, 2010), matrix completion (Candès & Recht, 2009) and matrix-factorisation recommenders (Koren et al., 2009). The new model corresponds to a cold-start user, the benchmarks to items, and the k benchmarks to the "interview items" of the item selection problem for cold-start recommendation (Meng et al., 2020). Missingness in leaderboards is not at random in Rubin's sense (Rubin, 1976) (benchmarks enter a leaderboard with its version, models with their release date), which we address with held-out-family and time-split evaluations. Psychometric treatments of leaderboards (Federiakin, 2025) are also related.

Benchmark suites used here. MTEB (Muennighoff et al., 2022) is our main data; HELM (Liang et al., 2022), the Open LLM Leaderboard and LiveBench provide small generative-LLM cross-checks (Section 5.1). MMLU (Hendrycks et al., 2020) underlies the HELM-MMLU subject matrix.

3 Problem and methods

Setting. Let $Y \in \mathbb{R}^{M\times N}$ be the historical score matrix of M models on N benchmarks with observed index set Ω (other cells are missing). A new model $m^\star$ has not been evaluated; it *can* be evaluated on a set $C \subseteq [N]$ of benchmarks. With budget k the evaluator chooses $S \subseteq C$, $|S| = k$, runs those benchmarks, and must infer the scores on a set of target benchmarks $T \subseteq [N] \setminus S$. The selection may use only Y (never the scores of $m^\star$), and a benchmark has to be one the model can actually run. Scores are standardised per benchmark with the mean and standard deviation of the *training* models, $z_{ij} = (y_{ij} - \mu_j)/s_j$, so that errors are comparable across benchmarks with different scales.

Inference rules (fixed S). All rules are fitted on training models only. *Factor model (`fa`).* $z_i = \mu + W f_i + \varepsilon_i$ with $f_i \sim \mathcal{N}(0, I_r)$, $\varepsilon_{ij} \sim \mathcal{N}(0, \psi_j)$, fitted by EM on the observed cells. Given z_S of the new model, the posterior mean of the factors is $\hat{f} = (I + W_S^\top \Psi_S^{-1} W_S)^{-1} W_S^\top \Psi_S^{-1}(z_S - \mu_S)$ and the prediction is $\hat{z} = \mu + W\hat{f}$; the rank r is chosen by hold-out error on 10% of the observed training cells. *Ridge (`ridge`).* Multi-output ridge regression of all benchmarks on the S columns of the factor-completed training matrix, with a GCV-chosen penalty. *Kernel ridge (`krr`).* Ridge plus an RBF kernel-ridge correction on the residuals (bandwidth by the median heuristic, penalty by GCV). *Nearest neighbours (`knn`):* mean of the 10 training models closest on S. *Average regression (`avgreg`):* a single aggregate (mean of z_S) mapped linearly to every target; this is the "one general-ability score" baseline. *Column mean (`mean`):* predict the training mean ($k = 0$ baseline). (An unregularised PCA projection was kept as a naive baseline on small matrices only.)

Selection rules. Every selector returns an *ordering* of benchmarks, computed from the training matrix; the subset for budget k is the first k entries that the new model can run, so subsets are nested in k and the same code serves any observed set C. Baselines: `random`; `popular` (most-evaluated first); `rawvar` (largest raw variance); `central` (largest mean absolute correlation with the others); `kcenter` (farthest-point on $1 - |\text{corr}|$). Spectral: `lev` (rank-r leverage scores) and `qr` (pivoted QR of the rank-r right singular vectors) (Mahoney & Drineas, 2009). Model-based greedy: `dopt` maximises $\log\det(I + \sum_{j\in S} w_j w_j^\top/\psi_j)$, the information about the latent factors; `gv_fa` greedily minimises the total conditional variance of all benchmarks given the chosen set under the factor covariance $WW^\top + \Psi$ (a pivoted Cholesky with trace-reduction pivoting; `gv_emp` uses a shrunk empirical covariance instead). A cost-aware variant divides the variance reduction by the benchmark's evaluation time (Appendix). Because the objective of `dopt` is monotone submodular, greedy attains $(1-1/e)$ of the optimum of *that* objective (Nemhauser et al., 1978); no such guarantee

is claimed for `gv_fa` on the squared error. An `oracle` selector (greedy on the new model's own target error) uses the targets and is reported only as an upper bound on what selection could give.

Cost and applicability. Fitting the factor model costs $O(n_{\text{iter}}\,(MNr^2 + Nr^3))$ on a training matrix; greedy selection costs $O(kN^2)$ from a covariance matrix; inference for a new model costs $O(kr^2+Nr)$ for `fa`. All of this runs in seconds on a CPU for the matrices here; the cost that matters to the evaluator is running the k benchmarks. The methods need a training matrix with enough models per benchmark (benchmarks with fewer than 8 observed training scores are dropped), a shared latent structure between benchmarks, and a new model drawn from a population the training models represent; Section 5 tests the last condition with held-out families and a time split.

Theory: what a rank-r Gaussian model predicts. Under the factor model with known parameters, the Bayes-optimal prediction given S has expected squared error on target j equal to $\psi_j + w_j^\top \Sigma_S w_j$ with posterior factor covariance $\Sigma_S = (I + \sum_{l\in S} w_l w_l^\top/\psi_l)^{-1}$. Consequently, selection only matters through Σ_S (it is the choice of well-spread, informative loadings), and for random S with i.i.d. loadings the excess error over the noise floor satisfies

$$\frac{rc}{1+kc} \;\le\; \frac{\mathbb{E}[\mathrm{MSE}] - \psi}{\psi} \;\le\; \frac{r}{k-r-1} \qquad (k > r+1), \tag{1}$$

with $c = \lambda/(r\psi)$ the per-factor signal-to-noise ratio (Proposition 1, Appendix A). Hence a random subset reaches a relative excess ϵ over the noise floor once $k \ge r + 1 + r/\epsilon$. We check Eq. (1) and the greedy-versus-optimum gap on synthetic matrices with known truth, and compare model-implied curves with measured ones (Section 5).

4 Evaluation protocol

The question is about a *new* model, so the unit of evaluation is a held-out model, not a held-out cell. The protocol was fixed before the final runs (study plan) and is as follows.

Splits. Models are split into folds (random 10-fold repeated a few times; leave-one-model-out as a check). Two shift splits test the "new model resembles old ones" assumption: *leave-family-out* (family = organisation for MTEB, base model for the Open LLM Leaderboard, so a family never appears on both sides) and a *temporal* split (train on the earliest 4/5 of dated models, test on the newest 1/5). Everything fitted is fitted on the training models of the fold: standardisation, factor model and rank, ridge/kernel penalties, *and the selection ordering*. For a held-out model the first k entries of the ordering it has scores for are revealed; models with fewer than k scores are skipped for that k.

Fixed targets (protocol B, primary). In each repetition a random 25% of the benchmarks (34% when $N \le 7$) are designated targets and are never selected; the error is measured on the held-out models' observed target cells only. Hence every k is scored on the same cells and curves over k are comparable. A secondary protocol A scores all remaining observed cells (the target set shrinks with k) and is used as a robustness check.

Metrics and uncertainty. The primary metric is nMSE, the mean over target cells of $((\hat{y}-y)/s_j)^2$ with s_j the training standard deviation, so that predicting the training mean ($k = 0$) scores about 1. We also report RMSE/MAE in score points (MTEB, HELM and LiveBench scores $\times 100$; Open LLM Leaderboard scores are already 0–100) and, for a subset of runs, Spearman correlation between predicted and measured scores across held-out models. All confidence intervals are 95% percentile bootstrap intervals over *held-out models* (500 resamples; model-level sums are first averaged over repetitions, folds and random-selector draws); comparisons between selectors or rules are paired on the same models and resamples. The random selector is averaged over 10 independent draws per fold (5 on the Open LLM Leaderboard). Seeds are fixed (`--seed 0`). These intervals treat held-out models as exchangeable and are not corrected for the many comparisons made; for the main MTEB and HELM-MMLU runs we also resample organisations and strict model families (Appendix B), which widens them modestly, and the variability across fold assignments is reported per repetition. Targets are one random draw per repetition (two to three on the MTEB blocks), so conclusions about unseen targets rest on few draws.

When is k "enough"? (fixed in advance) Let E_{mean} be the error of the column-mean predictor and E_{floor} the error of the same inference rule with *all* other observed benchmarks revealed (the full-information predictor, run on the same folds and targets). The gap-closed fraction is $G(k) = (E_{\text{mean}} - E_k)/(E_{\text{mean}} - E_{\text{floor}})$ and $k^\star(g)$ is the smallest k whose bootstrap lower confidence bound for G is at least $g \in \{0.8, 0.9\}$. Absolute tolerances ($\text{nMSE} \leq 0.5, 0.25, 0.15$; RMSE $\leq 3, 2$ points, upper bound of the CI) are reported as a secondary criterion. G can exceed 1 when the floor estimate is itself noisy.

Bounds and reference points. The full-information floor (not a method: it needs all benchmarks), the oracle selector (uses targets, optimistic), the column mean, random selection under the same rule, and where available a measurement-noise reference from repeated runs inside MTEB result files (a lower bound on irreducible error, available for only some tasks).

5 Experiments

5.1 Data

The main data are MTEB result files at the pinned commit (text-embedding models $\times$ embedding tasks; speed tasks and scores outside $[-1, 1]$ removed, test split preferred, `main_score` averaged over subsets, revisions of a model merged). The raw matrix is 12% observed and its missingness is tied to benchmark version and release date, so we evaluate on blocks that are nearly complete and treat density as an axis: `mteb-dense56` (tasks scored by $\geq$ 200 models), `mteb-dense` (275 tasks), `mteb-mid` (357) and `mteb-sparse` (472 tasks, 30% observed). The dense blocks are not a random sample: `mteb-dense56` keeps 28% of the models and 4% of the tasks of the raw leaderboard (10% of its observed cells; median release year 2024, median 245M parameters; Appendix B), so the results concern models that were run on a common core of tasks. HELM-MMLU (57 subjects), HELM Lite extended (91 models $\times$ 28 scenarios, built from the pinned HELM Lite v1.13.0 exports), HELM Lite (10 scenarios), LiveBench (7 categories) and the Open LLM Leaderboard (6 benchmarks) are generative-LLM matrices; each is analysed on its own (no cross-leaderboard name matching). The pinned Open LLM Leaderboard and LiveBench exports give only six and seven benchmarks, so these stay small-N (Appendix C). Nine real matrices are reported (four MTEB, five generative-LLM). Synthetic rank-r matrices with known truth are used for theory and condition checks only.

5.2 How accurate is inference from k benchmarks, and how do the rules compare?

Figure 1 gives the central result. Predicting the training mean gives $\text{nMSE} \approx 1.04$ (RMSE 15.0 score points) on MTEB-56. With $k = 4$ benchmarks run and kernel-ridge inference, the error falls to 0.30 for random selection and to 0.22 (greedy conditional variance) or 0.20 (greedy log-det), i.e. RMSE 7.8, 6.8 and 6.6 points; the full-information floor of the same rule is 0.15. The curves flatten around $k \approx$ 8–16 and the informed and random curves meet by $k \approx$ 16–32. On MTEB-275 the picture is the same with a higher floor (0.18 for the factor posterior, 0.32 for kernel ridge, which has more inputs than informative training models in the full-information case). The inference rule matters at least as much as selection (Table 2, Appendix; $k = 4$, random selection): the single-aggregate regression (`avgreg`) has 0.48 against 0.30 for kernel ridge, whereas kernel ridge with greedy log-det instead of random gains 0.10; the single-score baseline stays at 0.40–0.41 even at $k = 16$ (its floor is 0.40), so one "general ability" number cannot reproduce the other benchmarks here.

5.3 Does informed selection beat random selection?

Yes, at small k, and not beyond a moderate k (Fig. 2). On MTEB-56 greedy conditional-variance selection improves on random by 0.15 (95% CI 0.09–0.21) at $k = 1$–2, by 0.08 (0.05–0.11) at $k = 4$ and 0.04 at $k = 8$; at $k = 16$ the gain is not distinguishable from zero (-0.012, CI -0.034 to 0.008). Log-det selection gives -0.10 at $k = 4$ and remains significant at $k = 16$ (-0.030, CI -0.043 to -0.018) but not at $k = 32$. On MTEB-275 the gain at $k = 1$ is 0.24, falls to 0.04 at $k = 4$ (CI includes 0) and is slightly *negative* ($+0.014$) at $k = 32$; on HELM-MMLU it is 0.16 at $k = 1$ and

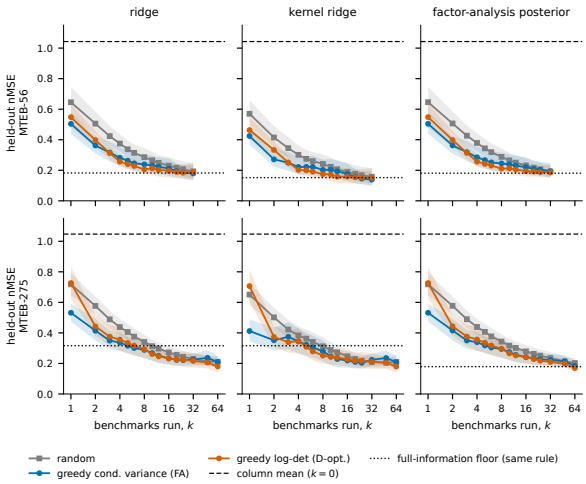


Figure 1: Held-out nMSE against the number k of benchmarks run (protocol B, random 10-fold CV, 95% bootstrap bands over held-out models) on MTEB-56 (152 models $\times$ 56 tasks, top) and MTEB-275 (133 $\times$ 275, bottom). Random selection and two greedy selectors fitted on training models, for three inference rules. Dashed: column-mean predictor; dotted: full-information floor of the same rule.

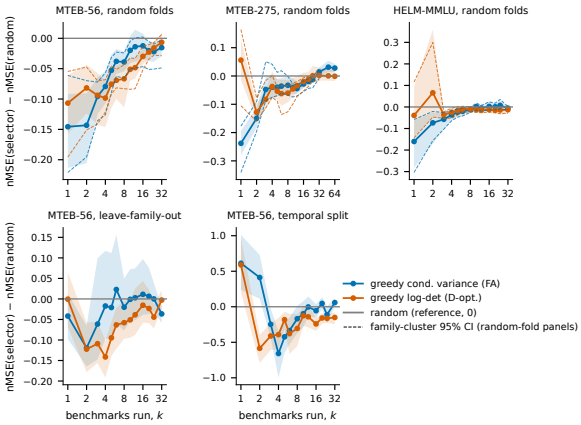


Figure 2: Paired difference in nMSE between an informed selector and random selection (kernel-ridge inference, same held-out models; below 0 favours the informed selector; paired 95% bootstrap bands over held-out models; dashed: strict-family bootstrap, random-fold panels). Random folds on three matrices, then leave-family-out and temporal splits (one fold of about 30 newest models) on MTEB-56.

0.04 at $k = 4$, and on the 91-model HELM Lite matrix 0.18, 0.11, 0.12, 0.05 at $k = 1, 2, 4, 8$. These intervals bootstrap models; resampling organisations or strict families widens the paired intervals by about 10–25% at $k \leq 2$ and keeps the sign of every difference up to $k = 8$ on MTEB-56 (families: $k = 1$ -0.146, CI -0.221 to -0.061; $k = 8$ -0.039, -0.066 to -0.024), whereas from $k = 12$ the interval contains 0. The benefit is largest where the budget is smallest, as the model-based theory predicts.

Not every selection heuristic helps: leverage-score, pivoted-QR and highest-variance selection, the standard column-subset-selection choices, are *worse* than random at $k \leq 8$ (at $k = 4$ with kernel ridge: 0.38, 0.37, 0.38 against 0.30 for random, Tables 2, 3), as is k-center (0.34). A diagnostic points to rank mismatch rather than redundancy: at the cross-validated rank their picks are *less* mutually correlated than random ones but have high task-specific variance (noisy tasks), and with the rank fixed at 3 they reach nMSE ≤ 0.5 at $k = 3$ instead of 4–6 (Appendix B). Picking the benchmarks most correlated with the others ("most central") behaves differently: it is as good as the

Table 1: Smallest k whose 95% bootstrap lower bound of the gap-closed fraction $G(k)$ reaches 0.8 and 0.9 (criterion fixed before the final runs; floor = same inference rule with all non-target benchmarks revealed, value in column "floor"; "rand.", "GV" = greedy conditional variance, "D-opt" = greedy log-det). krr: kernel ridge, fa: factor-analysis posterior. Grids: $k \in \{1, 2, \ldots, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64\}$ where N allows.

			$G \geq 0.8$			$G \geq 0.9$		
dataset	rule	floor	rand.	GV	D-opt	rand.	GV	D-opt
mteb-dense56	krr	0.15	5	2	3	12	8	4
mteb-dense56	fa	0.18	8	4	3	16	16	5
mteb-dense	krr	0.32	3	1	2	6	5	3
mteb-dense	fa	0.18	10	6	8	32	16	20
mteb-mid	krr	0.35	3	2	2	5	3	3
mteb-mid	fa	0.17	16	8	12	48	24	32
mteb-sparse	krr	0.43	3	2	4	5	3	4
mteb-sparse	fa	0.18	16	16	24	48	32	24
helm-mmlu	krr	0.28	2	1	3	3	1	3
helm-mmlu	fa	0.21	3	1	3	5	3	4
helm	krr	0.26	4	2	2	6	3	5
helm	fa	0.36	5	2	2	6	2	2
helm-lite-ext	krr	0.35	5	2	3	8	4	4
helm-lite-ext	fa	0.31	8	3	4	12	5	5
livebench	krr	0.34	1	1	1	2	2	2
livebench	fa	0.32	2	2	2	3	3	2
olb	krr	0.29	2	1	1	3	1	1
olb	fa	0.34	2	1	1	3	1	1

informed selectors at $k = 1$–2 on MTEB-56 (paired difference to random -0.13 and -0.09, against -0.15, -0.14 for conditional variance) and on MTEB-275 (-0.22, -0.12), is indistinguishable from random at $k = 4$ and $k = 16$, and on MTEB-275 and HELM-MMLU is *worse* than random at $k = 8$–16 ($+0.05$; $+0.04$). Direct paired contrasts (kernel ridge, family-clustered bootstrap; Appendix Figs. 11, 12, Tables 11–12) show that it is worse than conditional variance at $k = 2$–4 on MTEB-56 ($+0.06$; and against random only $k = 1$ stays separated from zero for $k \leq 8$ once families are resampled) and at $k = 8$–16 on MTEB-275, that conditional variance and log-det are not separated at $k \leq 4$ and that log-det is better from $k \geq 8$ ($+0.03$ for conditional variance minus log-det). Conditional variance is not the best deployable selector either: forward ridge selection of a shared set (`fwd_ridge`) and a greedy rule that re-estimates a local Gaussian after each revealed score (`adapt_loc`) are better by about 0.03–0.04 at $k = 8$–16 on MTEB-56,; none is uniformly best, and on HELM-MMLU the informed selectors are within 0.04 of each other at $k \geq 4$.

An oracle that picks benchmarks using the held-out model's own target scores (an optimistic upper bound; Table 6, Fig. 7a) reaches 0.29 and 0.23 at $k = 1, 2$ on MTEB-56 against 0.42/0.27 for greedy conditional variance and 0.57/0.42 for random, so a real gap to ideal selection remains at the smallest budgets and is small from $k = 4$. Fitting the selection model on the held-out rows as well (peeking) lowers the error of conditional-variance selection at $k = 4$ from 0.22 to 0.19 (Fig. 7b); selection is fitted on training models only everywhere else, so the reported gains are free of this optimism.

5.4 How large must k be?

Table 1 reports $k^\star$ under the pre-fixed criterion. On MTEB-56, $k^\star(0.8)$ is 2 for greedy conditional variance and 3 for log-det with kernel ridge, against 5 for random selection; $k^\star(0.9)$ is 8/4 against 12. With the factor posterior the corresponding numbers are 4/3 against 8 for $G \geq 0.8$ and 16/5 against 16 for $G \geq 0.9$. On the larger MTEB matrices more benchmarks are needed for the better-floored factor posterior ($k^\star(0.8) = 6$–12 for informed selection, 10 and 16 for random; $k^\star(0.9) = 16$–48), and on the small generative-LLM matrices ($N = 6$–57) $k^\star(0.8) \leq 4$ in all but one row — limited by N. $k^\star$ is relative to what the rule could achieve with everything revealed, and that floor is itself far from zero (0.15–0.43). With absolute tolerances (Appendix, Table 4) $\mathrm{nMSE} \leq 0.5$ needs $k \leq 5$ in all MTEB dense rows and $\mathrm{nMSE} \leq 0.25$ needs $k = 4$–64. $k^\star$ is a step function of noisy curves, so we report its spread (Appendix Tables 9, 10). Resampling organisations or strict families instead of models leaves $k^\star(0.8)$ unchanged for random (5) and log-det (3) and raises it for conditional variance ($2 \to 3$); its point value is 2 in all three repetitions against $4, 4, 4$ for random. Under shift the ranking among informed selectors is not stable (kernel ridge, MTEB-56, $G \geq 0.8$; conditional

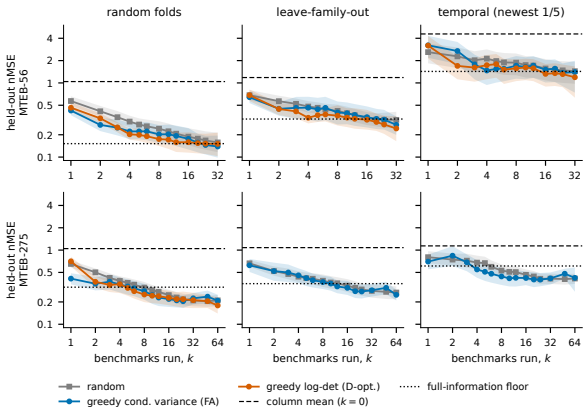


Figure 3: Held-out nMSE (kernel ridge, log axis) under random folds, leave-family-out and temporal splits on MTEB-56 (top) and MTEB-275 (bottom). The temporal split has one fold of about 30 test models; its intervals reflect those models only.

variance / log-det / random): random folds $2/3/5$, organisation leave-out $8/3/6$, strict-family leave-out $2/8/6$, temporal $4/2/6$. Under shift conditional variance beats random robustly only at $k \leq 3$. The answer to "how large k" is therefore: a handful of benchmarks (2–4 for 80% and 4–16 for 90% of the gap on MTEB-56, i.e. 4–29% of N) recovers most of the gain over the mean, but a few tens are needed to approach the accuracy that all other benchmarks would give, and there is no sharp elbow, in line with the intermediate effective rank of these matrices (Table 5, Appendix).

5.5 Data conditions and distribution shift

Missingness, M, rank (Appendix Fig. 10). Removing 30% or 60% of the training cells raises the error of greedy conditional variance with kernel ridge at $k = 4$ from 0.22 to 0.27 and 0.38 (random: 0.34, 0.48): selection still helps, but the error degrades. With $M = 60$, 30, 15 training models the error is 0.26, 0.46, 0.93 ($k = 4$); the training set must be several times larger than k. Fixing the factor rank at 2, 4 or 8 changes nMSE at $k = 16$ by at most 0.02.

A sparse leaderboard. On `mteb-sparse` (479×472, 30% observed; a new model reveals the first k benchmarks of the ordering that it has scores for, and models with fewer than k scores are skipped for that k) the column mean has nMSE 1.06 and the best full-information rule (factor posterior) 0.18. Conditional variance with the factor posterior gives $0.59/0.45/0.37/0.30$ at $k = 2/4/8/16$ against $0.65/0.52/0.42/0.33$ for random selection; the paired gain is separated from zero up to $k = 8$ (-0.083 at $k = 4$) but not at $k \geq 16$, and log-det is worse than random at $k \leq 4$. $k^\star(G \geq 0.8)$ with the factor posterior is 16 for conditional variance and random selection and 24 for log-det, so about 16 benchmarks are needed and informed selection no longer shortens it; kernel ridge reaches the criterion at $k = 2$–4 only because mean imputation makes its own floor poor (0.43; Appendix B).

Different matrices. All matrices have a dominant first component (36–80% of variance) but participation-ratio ranks of only 1.6–3.2 against hold-out-optimal factor ranks of 3–16 (Appendix Table 5); the small HELM-MMLU and Open LLM Leaderboard matrices hit their floors after $k \approx 3$–4. The gain from selection is clear on the Open LLM Leaderboard (0.16 at $k = 1$) and HELM Lite (0.16–0.19 at $k \leq 2$) but not distinguishable from zero on LiveBench at $k \leq 2$, and weak on MTEB-357 beyond $k = 8$. Selection conclusions thus hold at $k \leq 4$ across both families, but the generative-LLM curves are short.

Shift (Fig. 3). When whole families (organisations) are held out, the column-mean error rises to 1.18 and the full-information floors to 0.27–0.33 (from 0.15–0.18); at $k = 16$ kernel ridge reaches 0.33 (random) against ≈ 0.19 in random folds. Informed selection still helps at $k = 2$ (log-det -0.12, CI -0.18 to -0.06), at $k = 4$ only for log-det (-0.14; conditional variance -0.02, CI

-0.10 to 0.12), and not for $k \geq 16$. The temporal split is the clear failure case: the newest models lie outside the range of older ones, the column mean reaches 4.6 and even the full-information floor is 1.4–1.5, so no budget of $k \leq 32$ gives an error below 1.2 (log-det) on MTEB-56; on MTEB-275 the newest models are better predicted (0.42 at $k = 16$ against 1.14 for the mean). Neither remedy we tried changes this: a release-date covariate, or unshrunk least squares instead of the factor posterior, changes nMSE by at most 0.05 on the single temporal fold (MTEB-56, $k = 2/4/8/16$: posterior $1.96/1.50/1.56/1.49$, with trend $1.91/1.46/1.55/1.50$). In a rolling-origin evaluation (four cutoffs) the posterior with conditional-variance selection gives $0.96/0.91/0.87/0.74$ against 2.18 for the mean, and informed selection beats random at $k = 2$ (1.25) but not at $k = 16$ (Appendix Table 15). A Mahalanobis check on the k revealed scores under the training covariance (flag at the 95% χ^2 level) flags 7–8% of models in random folds, 5–15% under organisation or family leave-out and 39–48% of the temporal test models; flagged models have 3–$11\times$ the error of unflagged ones (kernel ridge, MTEB-56, random folds, $k = 16$: 0.94 against 0.12; temporal 2.5 against 0.75; Appendix Table 14). It is a warning, not a guarantee (on MTEB-275 it flags 1–3 of 15–27 temporal models). Conclusions about "a few benchmarks suffice" therefore apply to new models that resemble the training population and degrade, up to failure, for models that extend it; the flag rate at the chosen k should be reported with the result.

5.6 Does the theory predict the experiments?

Partly. On synthetic matrices the Monte Carlo excess error of random subsets lies inside the bounds of Proposition 1 in all 51 configurations (Fig. 8a–c, Appendix); greedy selection closes on average 0.87–1.00 of the random-to-optimal gap against brute force on $N = 16$, $r = 3$, but the worst case is poor (0.07 at $k = 2$), so the guarantee is for the information objective, not the error. The model-implied (Bayes) curves reproduce the ranking random $>$ informed and the saturation with k, and on synthetic data with known parameters the cross-validated curves to within 0.03 for the greedy rules. On real matrices the Bayes curve is lower than the held-out error, and fitting its covariance on the training models of each fold only leaves the gap: it is 20–30% below the cross-validated factor-posterior error on the MTEB matrices (median over k) and about 40% below on HELM-MMLU, for all three selection rules and with the same ordering of selectors (MTEB-56, log-det: 0.225 against 0.257 at $k = 4$, 0.135 against 0.194 at $k = 16$; Fig. 9). Theory thus explains the form of the k-curve and the advantage of informed selection at small k, not its level on real leaderboards.

6 Discussion and limitations

What the evidence supports. On score matrices that look like the training population, a few well-chosen benchmarks recover most of what the column mean misses: on MTEB-56, $k = 2$–4 informed benchmarks close 80% of the gap to the full-information predictor and $k = 4$–16 close 90% (5–8 and 12–16 for random ones; Table 1), and a single aggregate score does not (Table 2). The gain from *choosing* the benchmarks is largest at $k \leq 4$ and ≤ 0.03 nMSE from $k \approx 16$; the inference rule matters at least as much.

Failure cases. (i) *Extrapolation.* In the temporal split the newest MTEB-56 models lie outside the range of older ones: the column mean has $\mathrm{nMSE} = 4.6$, even the full-information floor is 1.4, and no $k \leq 32$ gives an error below 1.2; release-date terms and unshrunk least squares do not remove it. A Mahalanobis check on the k revealed scores flags these models (39–48% of the test models, with 3–$6\times$ the error of the unflagged ones; 7–8% of models in random folds, with 4–$8\times$), so we recommend reporting the flag rate; it is weak where the shift is mild (MTEB-275). Leave-family-out raises the floor from 0.15–0.18 to 0.27–0.33. (ii) *Spectral heuristics.* Leverage-score, pivoted-QR and highest-variance selection are worse than random at small k, apparently through rank mis-specification (Section 5). (iii) *Theory.* The Bayes curve stays 20–30% below the cross-validated error on MTEB even with a train-only covariance; it checks the shape of the curves, not their level.

Limits of the design. Targets are a random 25% of benchmarks, so prediction of a different *kind* of benchmark is not tested. The evidence is mostly MTEB: the five generative-LLM matrices have $N \leq 57$ (short k-curves); random folds are the optimistic reading of "new model". $k^\star$ is relative to an uncertain floor far from the noise level (0.03–0.06); the sparse result is one matrix.

AI-USE STATEMENT

This research was carried out end to end by a system of cooperating large-language-model agents (models of the Claude family), each holding one function: literature and data context, study planning and feasibility checking, analysis code and experiments, figures and tables, venue formatting, report writing, and independent internal and external review. The tasks for which ICLR requires disclosure were all done with generative AI: the research questions and hypotheses, the methodology, the theoretical derivations and proofs, the implementation, data cleaning and matrix construction, the generation of the synthetic matrices used for theory checks, the interpretation of results, and the drafting of this paper. The figures, code edits, literature search and polishing were also AI-assisted. No task was performed without generative AI assistance. No human wrote or edited text or code in the work reported here, and no human checked its content before this version; the verification that exists is mechanical and by other agents: every number, table and figure is generated by the shipped scripts from the pinned data, review agents independently read the code, the results and the draft against the task requirements, and cited works were looked up in public bibliographic records. These checks cannot exclude errors. No human author reviewed or takes responsibility for this work; the checks above are the only verification. This document is a research deliverable prepared in the ICLR 2027 format; it was not submitted to ICLR, and a real submission would require human authors who review the work and accept the responsibility that the ICLR AI policy places on authors.

REPRODUCIBILITY STATEMENT

All results come from the code shipped with the submission (`code/`, run from the repository root) on the pinned data versions listed in Appendix C. Seeds, the commands for each run group, environment versions, measured wall-clock times and the script that produces every table and figure are listed in the repository `README.md` and in `results/README.md`. All computation is CPU-only. Raw per-model result tables are not in git: they are published as the release `raw-results-1` of the experiment repository (compressed parts with checksums) and can be regenerated by the launch scripts; the summaries under `results/summary` are the inputs of every table and figure, and `code/check.sh` reproduces a small run end to end.

ETHICS STATEMENT

The study uses only public leaderboard results and synthetic matrices; no human subjects or personal data are involved. Several of the leaderboards used declare no data licence (Open LLM Leaderboard, LiveBench); we use their scores for research and redistribute only derived score matrices of small size. Predicting benchmark scores from a few evaluations could be misused to avoid evaluations that matter for safety or contamination checks; our results concern average score prediction on the leaderboards studied and say nothing about tail behaviour of individual models, and we recommend running the benchmarks that matter for a decision rather than inferring them.

REFERENCES

Emmanuel J. Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9:717–772, 2009.

Yifan Chen, Ethan N. Epperly, Joel A. Tropp, and Robert J. Webber. Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations. *Communications on Pure and Applied Mathematics*, 78:995–1041, 2025.

Abhimanyu Das and David Kempe. Submodular meets spectral: Greedy algorithms for subset selection, sparse approximation and dictionary selection. In *Proceedings of the 28th International Conference on Machine Learning*, 2011.

Denis Federiakin. Improving LLM leaderboards with psychometrical methodology. *arXiv preprint arXiv:2501.17200*, 2025.

Carlos Guestrin, Andreas Krause, and Ajit Paul Singh. Near-optimal sensor placements in Gaussian processes. In *Proceedings of the 22nd International Conference on Machine Learning*, pp. 265–272, 2005.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Alexander Ilin and Tapani Raiko. Practical approaches to principal component analysis in the presence of missing values. *Journal of Machine Learning Research*, 11:1957–2000, 2010.

Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz. metabench – a sparse benchmark of reasoning and knowledge in large language models. *arXiv preprint arXiv:2407.12844*, 2024.

Yehuda Koren, Robert Bell, and Chris Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37, 2009.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Michael W. Mahoney and Petros Drineas. CUR matrix decompositions for improved data analysis. *Proceedings of the National Academy of Sciences*, 106(3):697–702, 2009.

Yitong Meng, Jie Liu, Xiao Yan, and James Cheng. The item selection problem for user cold-start recommendation. *arXiv preprint arXiv:2010.14013*, 2020.

Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. *arXiv preprint arXiv:2210.07316*, 2022.

George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14:265–294, 1978.

Yotam Perlitz, Ariel Gera, Ofir Arviv, Asaf Yehudai, Elron Bandel, Eyal Shnarch, Michal Shmueli-Scheuer, and Leshem Choshen. Do these LLM benchmarks agree? fixing benchmark evaluation with BenchBench. *arXiv preprint arXiv:2407.13696*, 2024.

Felipe Maia Polo, Seamus Somerstep, Leshem Choshen, Yuekai Sun, and Mikhail Yurochkin. Sloth: scaling laws for LLM skills to predict multi-benchmark performance across families. *arXiv preprint arXiv:2412.06540*, 2024a.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: evaluating LLMs with fewer examples. *arXiv preprint arXiv:2402.14992*, 2024b.

Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. *arXiv preprint arXiv:2405.10938*, 2024.

Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976.

A THEORY: DERIVATIONS

Throughout, $z = Wf + \varepsilon$ with $f \sim \mathcal{N}(0, I_r)$, $\varepsilon \sim \mathcal{N}(0, \Psi)$, $\Psi = \mathrm{diag}(\psi)$, $W \in \mathbb{R}^{N\times r}$ with rows w_j, and parameters are known. For an observed set S the posterior of f given z_S is Gaussian with covariance $\Sigma_S = (I + \sum_{l\in S} w_l w_l^\top/\psi_l)^{-1}$ and the posterior mean of z_j, $j \notin S$, has expected squared error $\psi_j + w_j^\top \Sigma_S w_j$ (it does not depend on the observed values).

Proposition 1 (Random subsets)**.** *Let $\psi_j \equiv \psi$ and let the rows w_j be i.i.d. $\mathcal{N}(0, (\lambda/r)I_r)$; put $c = \lambda/(r\psi)$. For a uniformly random S with $|S| = k$ and a target $j \notin S$, the relative excess error $e(k) = (\mathbb{E}\,\mathrm{MSE}_j - \psi)/\psi$ satisfies*

$$\frac{rc}{1+kc} \le e(k), \qquad e(k) \le \frac{r}{k-r-1} \;\; (k > r+1).$$

Proof. w_j is independent of S, so $\mathbb{E}[w_j^\top \Sigma_S w_j] = (\lambda/r)\,\mathbb{E}\,\mathrm{tr}\,\Sigma_S$ and $e(k) = c\,\mathbb{E}\,\mathrm{tr}\,\Sigma_S$. Write $A = \sum_{l\in S} w_l w_l^\top/\psi = cQ$ with $Q \sim \mathrm{Wishart}_r(k, I)$, so $\mathbb{E}A = kcI$. The map $A \mapsto \mathrm{tr}(I+A)^{-1}$ is convex on positive semidefinite matrices, so Jensen gives $\mathbb{E}\,\mathrm{tr}(I+A)^{-1} \geq \mathrm{tr}(I+kcI)^{-1} = r/(1+kc)$, hence the lower bound. For the upper bound, $(I+A)^{-1} \preceq A^{-1}$ when $A \succ 0$ ($k \geq r$), and $\mathbb{E}\,\mathrm{tr}\,Q^{-1} = r/(k-r-1)$ for $k > r+1$, so $\mathbb{E}\,\mathrm{tr}\,\Sigma_S \leq r/(c(k-r-1))$ and $e(k) \leq r/(k-r-1)$. □

Consequently a necessary condition for $e(k) \leq \epsilon$ is $k \geq r/\epsilon - 1/c$ and a sufficient one is $k \geq r + 1 + r/\epsilon$.

Proposition 2 (Greedy information selection). *$F(S) = \log\det\left(I + \sum_{l\in S} v_l v_l^\top\right)$ with $v_l = w_l/\sqrt{\psi_l}$ is monotone and submodular, so greedy selection of k benchmarks attains at least $(1-1/e)$ of $\max_{|S|=k} F(S)$.*

Proof. With $A_S = I + \sum_{l\in S} v_l v_l^\top$ the marginal gain of j is $F(S\cup j) - F(S) = \log(1 + v_j^\top A_S^{-1} v_j) \geq 0$. If $S \subseteq S'$ then $A_S \preceq A_{S'}$, so $A_S^{-1} \succeq A_{S'}^{-1}$ and the gain is smaller for S': submodularity. The guarantee is the classical one for monotone submodular maximisation under a cardinality constraint (Nemhauser et al., 1978). □

$F(S) = 2\,I(f; z_S)$ is the mutual information between the factors and the observations, which is why `dopt` targets the latent factors rather than the squared error of the targets. The error-driven rule `gv_fa` minimises $\sum_j \mathrm{Var}(z_j \mid z_S)$ greedily; it has no submodularity guarantee in general, and its quality relative to the optimum is therefore checked by brute force on small instances (Section 5).

B Additional results

Selector × inference rule. Table 2 gives nMSE for six selectors and five inference rules on MTEB-56 at $k = 2, 4, 8, 16$, and Table 3 for all ten selectors (ridge and kernel ridge use the same completed training matrix; intervals are 95% bootstrap half-widths and overlap in most cells, so compare selectors in the paired Fig. 2). Figure 4 shows the curves for every selector.

Table 2: nMSE (± half-width of the 95% bootstrap CI over held-out models) for six selectors and five inference rules on MTEB-56 (protocol B, random 10-fold CV), at $k = 2, 4, 8, 16$. Bold: lowest point estimate in the column within each k.

selector	avg-reg.	kNN	ridge	KRR	FA
$k = 2$					
random	0.55 ±0.08	0.41 ±0.06	0.51 ±0.07	0.41 ±0.07	0.51 ±0.07
greedy cond. variance (FA)	**0.45 ±0.06**	**0.28 ±0.05**	**0.36 ±0.06**	**0.27 ±0.06**	**0.36 ±0.06**
greedy log-det (D-opt.)	0.47 ±0.06	0.34 ±0.05	0.40 ±0.05	0.33 ±0.05	0.40 ±0.05
k-center	0.66 ±0.11	0.52 ±0.08	0.57 ±0.09	0.50 ±0.08	0.57 ±0.09
leverage score	0.74 ±0.12	0.65 ±0.10	0.71 ±0.11	0.63 ±0.11	0.71 ±0.11
most central	0.46 ±0.07	0.34 ±0.05	0.42 ±0.06	0.33 ±0.05	0.42 ±0.06
$k = 4$					
random	0.48 ±0.07	0.31 ±0.05	0.37 ±0.06	0.30 ±0.06	0.38 ±0.06
greedy cond. variance (FA)	**0.41 ±0.06**	0.25 ±0.05	0.28 ±0.06	0.22 ±0.06	0.28 ±0.06
greedy log-det (D-opt.)	0.41 ±0.06	**0.24 ±0.04**	**0.26 ±0.04**	**0.20 ±0.03**	**0.26 ±0.04**
k-center	0.54 ±0.09	0.34 ±0.05	0.40 ±0.05	0.34 ±0.07	0.40 ±0.05
leverage score	0.57 ±0.08	0.41 ±0.06	0.49 ±0.06	0.38 ±0.06	0.49 ±0.06
most central	0.44 ±0.06	0.29 ±0.05	0.36 ±0.06	0.28 ±0.05	0.36 ±0.06
$k = 8$					
random	0.43 ±0.06	0.26 ±0.05	0.29 ±0.05	0.24 ±0.06	0.29 ±0.05
greedy cond. variance (FA)	0.43 ±0.06	0.24 ±0.05	0.24 ±0.06	0.20 ±0.06	0.24 ±0.06
greedy log-det (D-opt.)	**0.40 ±0.05**	**0.22 ±0.04**	**0.21 ±0.03**	**0.17 ±0.04**	**0.21 ±0.04**
k-center	0.46 ±0.06	0.29 ±0.05	0.31 ±0.05	0.24 ±0.05	0.32 ±0.05
leverage score	0.47 ±0.07	0.30 ±0.05	0.33 ±0.04	0.24 ±0.04	0.33 ±0.04
most central	0.41 ±0.05	0.24 ±0.04	0.24 ±0.04	0.20 ±0.04	0.25 ±0.04
$k = 16$					
random	0.41 ±0.06	0.23 ±0.04	0.23 ±0.05	0.19 ±0.05	0.23 ±0.04
greedy cond. variance (FA)	0.41 ±0.06	0.24 ±0.04	0.21 ±0.06	0.18 ±0.06	0.22 ±0.06
greedy log-det (D-opt.)	0.40 ±0.06	**0.21 ±0.04**	**0.20 ±0.05**	**0.16 ±0.05**	**0.19 ±0.04**
k-center	0.43 ±0.06	0.25 ±0.04	0.23 ±0.04	0.17 ±0.04	0.24 ±0.04
leverage score	0.42 ±0.06	0.25 ±0.04	0.23 ±0.03	0.17 ±0.03	0.24 ±0.03
most central	**0.40 ±0.06**	0.22 ±0.04	0.20 ±0.04	0.17 ±0.04	0.22 ±0.04

Table 3: As Table 2 for all ten selectors at $k = 2, 4, 8, 16$ (adds highest variance, most covered and conditional variance under the empirical covariance).

selector	avg-reg.	kNN	ridge	KRR	FA
$k = 2$					
random	0.55 ±0.08	0.41 ±0.06	0.51 ±0.07	0.41 ±0.07	0.51 ±0.07
greedy cond. variance (FA)	**0.45 ±0.06**	**0.28 ±0.05**	**0.36 ±0.06**	**0.27 ±0.06**	**0.36 ±0.06**
greedy log-det (D-opt.)	0.47 ±0.06	0.34 ±0.05	0.40 ±0.05	0.33 ±0.05	0.40 ±0.05
greedy cond. variance (empirical)	0.46 ±0.06	0.29 ±0.05	0.37 ±0.06	0.28 ±0.06	0.37 ±0.06
k-center	0.66 ±0.11	0.52 ±0.08	0.57 ±0.09	0.50 ±0.08	0.57 ±0.09
leverage score	0.74 ±0.12	0.65 ±0.10	0.71 ±0.11	0.63 ±0.11	0.71 ±0.11
pivoted QR	0.73 ±0.12	0.64 ±0.10	0.69 ±0.11	0.63 ±0.10	0.69 ±0.11
most central	0.46 ±0.07	0.34 ±0.05	0.42 ±0.06	0.33 ±0.05	0.42 ±0.06
highest variance	0.62 ±0.11	0.45 ±0.09	0.60 ±0.10	0.45 ±0.09	0.60 ±0.10
most covered	0.59 ±0.10	0.41 ±0.08	0.56 ±0.10	0.42 ±0.07	0.56 ±0.10
$k = 4$					
random	0.48 ±0.07	0.31 ±0.05	0.37 ±0.06	0.30 ±0.06	0.38 ±0.06
greedy cond. variance (FA)	**0.41 ±0.06**	0.25 ±0.05	0.28 ±0.06	0.22 ±0.06	0.28 ±0.06
greedy log-det (D-opt.)	0.41 ±0.06	**0.24 ±0.04**	**0.26 ±0.04**	**0.20 ±0.03**	**0.26 ±0.04**
greedy cond. variance (empirical)	0.41 ±0.06	0.25 ±0.05	0.28 ±0.06	0.23 ±0.06	0.29 ±0.06
k-center	0.54 ±0.09	0.34 ±0.05	0.40 ±0.05	0.34 ±0.07	0.40 ±0.05
leverage score	0.57 ±0.08	0.41 ±0.06	0.49 ±0.06	0.38 ±0.06	0.49 ±0.06
pivoted QR	0.55 ±0.09	0.39 ±0.05	0.47 ±0.05	0.37 ±0.06	0.47 ±0.05
most central	0.44 ±0.06	0.29 ±0.05	0.36 ±0.06	0.28 ±0.05	0.36 ±0.06
highest variance	0.59 ±0.10	0.39 ±0.07	0.51 ±0.07	0.38 ±0.08	0.51 ±0.07
most covered	0.49 ±0.08	0.33 ±0.06	0.38 ±0.06	0.32 ±0.06	0.39 ±0.06
$k = 8$					
random	0.43 ±0.06	0.26 ±0.05	0.29 ±0.05	0.24 ±0.06	0.29 ±0.05
greedy cond. variance (FA)	0.43 ±0.06	0.24 ±0.05	0.24 ±0.06	0.20 ±0.06	0.24 ±0.06
greedy log-det (D-opt.)	**0.40 ±0.05**	**0.22 ±0.04**	**0.21 ±0.03**	**0.17 ±0.04**	**0.21 ±0.04**
greedy cond. variance (empirical)	0.43 ±0.06	0.24 ±0.05	0.24 ±0.06	0.21 ±0.06	0.25 ±0.06
k-center	0.46 ±0.06	0.29 ±0.05	0.31 ±0.05	0.24 ±0.05	0.32 ±0.05
leverage score	0.47 ±0.07	0.30 ±0.05	0.33 ±0.04	0.24 ±0.04	0.33 ±0.04
pivoted QR	0.45 ±0.06	0.29 ±0.05	0.32 ±0.04	0.24 ±0.04	0.32 ±0.04
most central	0.41 ±0.05	0.24 ±0.04	0.24 ±0.04	0.20 ±0.04	0.25 ±0.04
highest variance	0.51 ±0.08	0.29 ±0.05	0.32 ±0.06	0.25 ±0.06	0.32 ±0.05
most covered	0.42 ±0.06	0.27 ±0.04	0.27 ±0.04	0.21 ±0.03	0.27 ±0.04
$k = 16$					
random	0.41 ±0.06	0.23 ±0.04	0.23 ±0.05	0.19 ±0.05	0.23 ±0.04
greedy cond. variance (FA)	0.41 ±0.06	0.24 ±0.04	0.21 ±0.06	0.18 ±0.06	0.22 ±0.06
greedy log-det (D-opt.)	0.40 ±0.06	**0.21 ±0.04**	**0.20 ±0.05**	**0.16 ±0.05**	**0.19 ±0.04**
greedy cond. variance (empirical)	0.41 ±0.06	0.23 ±0.04	0.21 ±0.05	0.17 ±0.05	0.22 ±0.05
k-center	0.43 ±0.06	0.25 ±0.04	0.23 ±0.04	0.17 ±0.04	0.24 ±0.04
leverage score	0.42 ±0.06	0.25 ±0.04	0.23 ±0.03	0.17 ±0.03	0.24 ±0.03
pivoted QR	0.42 ±0.06	0.24 ±0.04	0.22 ±0.03	0.17 ±0.03	0.22 ±0.03
most central	**0.40 ±0.06**	0.22 ±0.04	0.20 ±0.04	0.17 ±0.04	0.22 ±0.04
highest variance	0.47 ±0.07	0.25 ±0.05	0.26 ±0.05	0.21 ±0.06	0.26 ±0.05
most covered	0.42 ±0.06	0.23 ±0.05	0.21 ±0.04	0.18 ±0.04	0.21 ±0.03

$k^\star$ with other floors and absolute tolerances. Table 4 repeats Table 1 with the floor of the best inference rule instead of the same rule (G_{best}), and adds the smallest k at which the upper confidence bound of nMSE is below an absolute tolerance. "$>$grid": not reached for any k on the grid.

Matrices.

Oracle, leakage and cost. Table 6 compares random, the two deployable greedy selectors and the oracle (which uses the held-out model's own target scores and is therefore an optimistic bound) on four matrices. The oracle is clearly better only at $k \leq 2$ on MTEB-56 (0.23 against 0.27 and 0.33 at $k = 2$) and on HELM Lite and the Open LLM Leaderboard ($k = 2$: 0.19 and 0.22 against 0.31–0.36); at $k = 4$ and 8 on MTEB-56 it is within the confidence intervals of log-det selection. Table 7 lists the leakage check of Fig. 7b by k. Cost-aware selection (greedy conditional variance per unit cost, and a cheapest-first rule) reaches $\mathrm{nMSE} = 0.26$ with kernel ridge at a budget of $5‰$ of the total evaluation time of the 56 tasks, against 0.49 for random tasks and 0.42 for cost-blind conditional variance, which pays for expensive tasks first; at $20‰$ the values are 0.19 (cost-aware), 0.40 (random) and 0.42 (cost-blind). This holds on MTEB-56 only: on MTEB-275 the cost-aware rules are good at 5–$10‰$ (0.20–0.23 against 0.26–0.37 for cost-blind and 0.34–0.39 for random) but at $20‰$ they are *worse than random* (0.75 and 1.24 against 0.29), and from $160‰$ they stay above both other rules (0.34 against 0.26 and 0.25) until the budget reaches $640‰$ (0.32 against 0.56 and

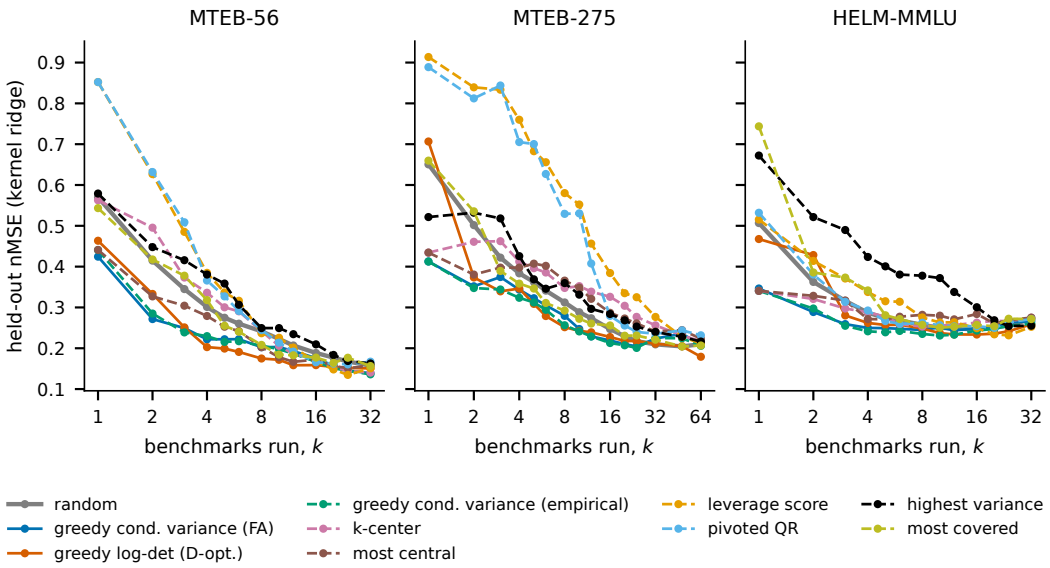


Figure 4: Held-out nMSE against k for all ten selectors (kernel ridge, protocol B) on MTEB-56, MTEB-275 and HELM-MMLU; bands are 95% bootstrap CIs over held-out models. Leverage-score, pivoted-QR and highest-variance selection are worse than random at small k.

Table 4: Smallest k meeting $G_{\text{best}} \geq 0.8$, 0.9 (95% lower bound; floor = best rule) and absolute tolerances nMSE ≤ 0.5, 0.25 (95% upper bound). Abbreviations as in Table 1.

			$G_{\text{best}} \geq 0.8$			$G_{\text{best}} \geq 0.9$			nMSE≤ 0.5			nMSE≤ 0.25		
dataset	rule	floor	rand.	GV	D-opt	rand.	GV	D-opt	rand.	GV	D-opt	rand.	GV	D-opt
mteb-dense56	krr	0.15	5	2	3	12	8	4	2	1	2	16	16	4
mteb-dense56	fa	0.18	8	5	4	24	20	10	3	2	2	32	32	16
mteb-dense	krr	0.32	8	8	6	20	12	16	3	1	2	32	24	64
mteb-dense	fa	0.18	10	6	8	32	16	20	4	2	3	48	64	32
mteb-mid	krr	0.35	10	6	6	20	16	16	3	2	2	>grid	24	24
mteb-mid	fa	0.17	16	8	12	48	24	32	5	3	5	64	48	48
mteb-sparse	krr	0.43	10	12	12	48	>grid	>grid	5	2	4	>grid	>grid	>grid
mteb-sparse	fa	0.18	16	16	24	48	32	24	6	5	8	48	>grid	>grid
helm-mmlu	krr	0.28	2	1	3	4	2	3	4	2	3	>grid	>grid	>grid
helm-mmlu	fa	0.21	3	1	3	5	3	4	4	3	4	>grid	>grid	>grid
helm	krr	0.26	4	2	2	6	3	5	4	2	2	>grid	>grid	>grid
helm	fa	0.36	>grid	4	5	>grid	>grid	>grid	>grid	2	2	>grid	>grid	>grid
helm-lite-ext	krr	0.35	6	3	4	12	4	4	8	3	4	>grid	>grid	>grid
helm-lite-ext	fa	0.31	8	3	4	12	5	5	10	3	4	>grid	>grid	>grid
livebench	krr	0.34	2	2	1	2	4	2	1	1	1	>grid	>grid	>grid
livebench	fa	0.32	2	2	2	3	3	2	2	2	2	>grid	>grid	>grid
olb	krr	0.29	2	1	1	>grid	>grid	>grid	2	1	1	>grid	>grid	>grid
olb	fa	0.34	3	2	2	>grid	>grid	>grid	2	1	1	>grid	>grid	>grid

0.43). We therefore do not claim a general result for cost-aware selection and did not investigate the cause. Task costs are the median evaluation time per task from the result files, a coarse proxy.

Synthetic data conditions. Table 8 shows selection on synthetic rank-r matrices with known structure (factor-analysis inference; these are controlled checks and not evidence about any real leaderboard). Informed selection beats random selection in every condition at $k \leq 4$ (e.g. 0.25 and 0.23 against 0.40 on the base matrix at $k = 4$); the gain is smaller with high noise (0.66 $\rightarrow$ 0.51, $k = 4$), with rank 8 (0.68 $\rightarrow$ 0.58–0.63, $k = 4$), and with correlated benchmark clusters at $k = 8$ (0.42 $\rightarrow$ 0.38–0.40), and with $M = 30$ training models the errors are higher throughout.

Missingness, training-set size and rank. Fig. 10 shows the sensitivity analyses described in Section 5.

Dependence-robust intervals and $k^\star$ by split. The main intervals bootstrap held-out models. We also resampled organisations and strict model families (union of organisations that share a model

Table 5: Score matrices (models M × benchmarks N). *PR rank*: participation-ratio effective rank of the standardised matrix; *FA rank (CV)*: rank chosen by hold-out error; *noise nMSE*: mean measurement-noise variance on the cells with repeated runs, in units of the benchmark variance (a lower bound on unavoidable error, available only for the 18% (28%) of cells of MTEB-56 (MTEB-275) with repeat runs). Generated by `code/figtab/tables.py` from `results/summary/datasets.csv`.

dataset	M	N	obs. %	PC1 %	PCs 80/90%	PR rank	FA rank (CV)	noise nMSE
mteb-dense56	152	56	99.6	63	3 / 9	2.4	12	0.028
mteb-dense	133	275	98.9	54	7 / 17	3.2	16	0.060
mteb-mid	156	357	92.1	54	8 / 19	3.2	24	–
mteb-sparse	479	472	30.2	36	19 / 43	6.8	16	–
helm-mmlu	79	57	100.0	80	2 / 4	1.6	3	–
helm	91	10	99.9	63	2 / 4	2.3	3	–
helm-lite-ext	91	28	99.9	61	4 / 8	2.6	6	–
livebench	195	7	76.7	62	3 / 4	2.4	2	–
olb	4497	6	100.0	74	2 / 3	1.8	2	–

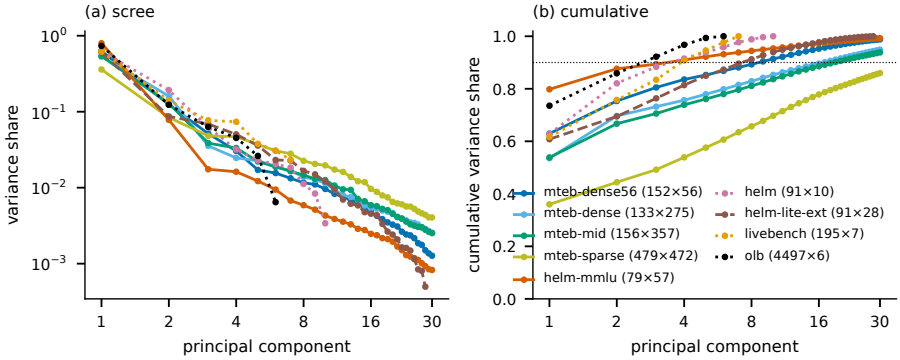


Figure 5: Spectrum of each standardised real score matrix (PCA on the completed matrix): (a) variance share per principal component, (b) cumulative share, dotted line 90%.

family; the HELM organisation is a name-derived proxy), 1000 resamples, on the three MTEB blocks and HELM-MMLU (`cluster_*.csv`). Fig. 11 compares the two intervals for the paired contrasts; family-clustered intervals are 0.46–2.53 times as wide as the model-level ones (median 1.11 over the plotted contrasts), keep the sign of conditional variance and log-det against random almost everywhere and weaken the case for centrality against random (on MTEB-56 only $k = 1$ remains separated from zero for $k \leq 8$; $k = 2$: $[-0.133, 0.003]$, $k = 8$: $[-0.066, 0.015]$). Table 9 gives $k^\star$ under both schemes and its variation across repetitions (the entry for conditional variance with the factor posterior at $g = 0.9$ is 20 here and 16 in Table 1 on MTEB-56 and MTEB-275: the lower bound of G grazes 0.9 over several k, so this value moves with the resampling seed; random with the factor posterior at $g = 0.9$ on MTEB-357 is ">grid" here, the grid ending at 32, against 48 in Table 1); Table 10 gives $k^\star$ under organisation, strict-family and temporal splits. Leave-organisation-out leaves no held-out model with two same-organisation models in training, but 47% (MTEB-56) and 52% (MTEB-275) still have two same-family models; strict-family leave-out removes both (`group_counts.csv`).

Extra selectors and rules. Fig. 12 (top) compares, on the same folds and targets, random, central, conditional variance, log-det, `fwd_ridge` (forward selection of one shared benchmark set by cross-validated ridge error on the training models), `mtl` (multi-task lasso path) and `adapt_loc` (greedy conditional variance under a local Gaussian: the 40 training models nearest to the new model on the scores revealed so far, shrunk 10% to the diagonal, re-estimated after each revealed score; with one global Gaussian the posterior variance does not depend on the revealed values, so an adaptive rule equals the static one). On MTEB-56 with the factor posterior at $k = 8$ the errors are 0.20 (`adapt_loc`), 0.20 (`fwd_ridge`), 0.21 (log-det), 0.22 (`mtl`), 0.25 (conditional variance), 0.25 (central), 0.29 (random). On rules (bottom): unshrunk least squares (`ols`) is close to ridge

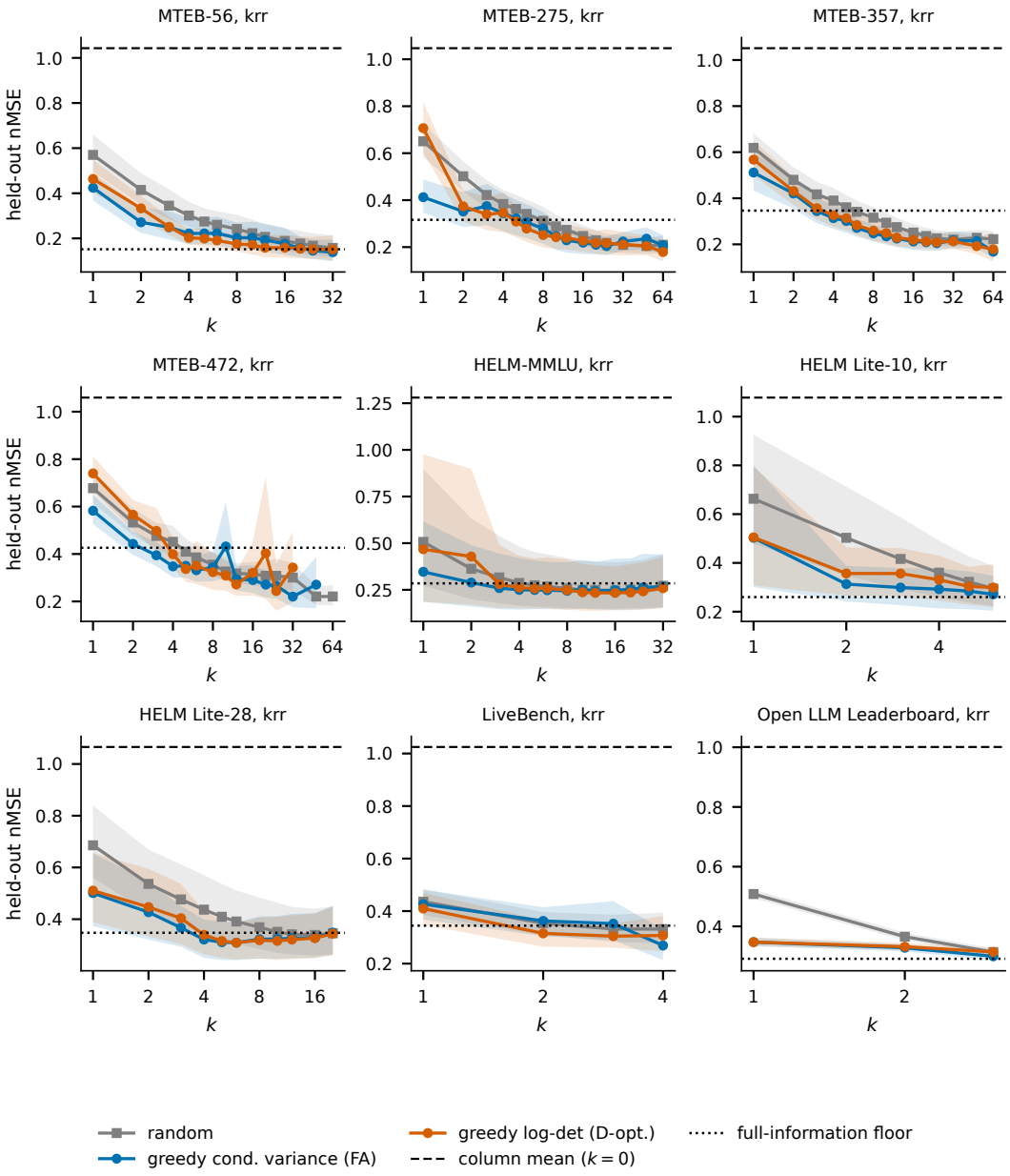


Figure 6: The comparison of Fig. 1 on every real matrix (protocol B, random folds): random selection against greedy conditional variance and greedy log-det; k axes differ because N ranges from 6 to 472.

and the factor posterior; a one-factor (IRT-style) posterior is clearly worse (MTEB-56, conditional-variance set, $k = 8$: 0.40 against 0.25), a rank-2 posterior is in between (0.31) and soft-impute is worse than the factor posterior at small k (0.31 at $k = 8$). Per-target forward selection (a separate set for each target, at matched union size) is no better than a shared set with the same rule (`extra_pertarget_summary.csv`; MTEB-56, unions of 18–23 benchmarks: 0.200 against 0.210 for the factor posterior on the union, and 0.172 for kernel ridge on the union); on the larger blocks it is worse (MTEB-275, unions of 50–84: 0.202 against 0.165; MTEB-357, 63–116: 0.163 against 0.116).

Leverage and QR at fixed rank. With the rank of the leverage and QR criteria fixed at 3 rather than chosen by cross-validation (8–16), both reach nMSE $\leq$ 0.5 at $k = 3$

Table 6: Oracle selection against random and deployable greedy selectors (kernel ridge; nMSE ± half-width of the 95% CI; GV: greedy conditional variance; "–": k outside the dataset's grid).

dataset	k	random	GV	D-opt	oracle
mteb-dense56	2	0.41 ±0.07	0.27 ±0.06	0.33 ±0.05	0.23 ±0.03
	4	0.30 ±0.06	0.22 ±0.06	0.20 ±0.03	0.19 ±0.03
	8	0.24 ±0.06	0.20 ±0.06	0.17 ±0.04	0.15 ±0.03
helm-mmlu	2	0.36 ±0.22	0.29 ±0.16	0.43 ±0.36	0.29 ±0.18
	4	0.29 ±0.16	0.25 ±0.14	0.26 ±0.14	0.22 ±0.12
	8	0.26 ±0.13	0.24 ±0.13	0.25 ±0.13	0.18 ±0.10
helm	2	0.50 ±0.18	0.31 ±0.07	0.36 ±0.10	0.19 ±0.05
	4	0.36 ±0.12	0.29 ±0.08	0.33 ±0.09	0.18 ±0.06
	8	–	–	–	–
livebench	2	0.35 ±0.04	0.36 ±0.06	0.32 ±0.05	0.21 ±0.03
	4	0.33 ±0.06	0.27 ±0.06	0.31 ±0.07	0.31 ±0.06
	8	–	–	–	–
olb	2	0.36 ±0.01	0.32 ±0.01	0.32 ±0.02	0.22 ±0.01
	4	–	–	–	–
	8	–	–	–	–

Table 7: Leakage check (kernel ridge, nMSE): selection model fitted on training models only / also on the held-out rows. Random selection is identical by construction.

dataset	selector	$k = 2$	$k = 4$	$k = 8$	$k = 16$
mteb-dense56	random	0.415 / 0.415	0.301 / 0.301	0.241 / 0.241	0.188 / 0.188
mteb-dense56	greedy cond. variance (FA)	0.271 / 0.269	0.222 / 0.190	0.203 / 0.195	0.176 / 0.168
mteb-dense56	greedy log-det (D-opt.)	0.333 / 0.334	0.203 / 0.204	0.175 / 0.166	0.159 / 0.131
helm-mmlu	random	0.363 / 0.363	0.287 / 0.287	0.257 / 0.257	0.246 / 0.246
helm-mmlu	greedy cond. variance (FA)	0.289 / 0.297	0.250 / 0.238	0.245 / 0.236	0.249 / 0.244
helm-mmlu	greedy log-det (D-opt.)	0.428 / 0.330	0.262 / 0.236	0.248 / 0.228	0.234 / 0.234

(kernel ridge) against 4–6 at the cross-validated rank, and nMSE $\leq$ 0.25 at $k = 12$ against 12–16 (`kstar_d56_selrank2.csv`); at the cross-validated rank their first eight picks have mean pairwise $|\text{corr}|$ 0.43–0.49 (random 0.62, conditional variance 0.59, log-det 0.67) and mean uniqueness 0.24–0.29 (conditional variance and random 0.09–0.10, log-det 0.01) (`lev_diag_mteb_dense56.csv`). The failure of these criteria at small k is thus mostly a rank mis-specification that selects task-specific (noisy) tasks, not a redundant cluster.

Temporal drift and the out-of-range check. Fig. 12 (bottom) and Table 15 show the rules on the single temporal fold and under rolling origin, Table 14 the detector. The detector scores the k revealed scores of a new model by their squared Mahalanobis distance under the shrunk empirical covariance of the training models and flags it above the 95% χ^2 quantile; it is a check on the input, not a calibrated predictor of error. In a rolling-origin evaluation (four cutoffs at 55–85% of dated models, next 15% as test) the factor posterior with conditional-variance selection gives $0.96/0.91/0.87/0.74$ at $k = 2/4/8/16$ on MTEB-56 against 2.18 for the mean and 2.03 for the mean with trend; on MTEB-275 $0.58/0.48/0.45/0.39$ against 1.28 and 1.37 (`extra_temporal_summary.csv`).

Sparse matrix. `mteb-sparse` has 479 models × 472 tasks with 30.2% of the cells observed (2 repetitions of 10-fold CV, about 1.2 h each); its description in `sparse_population.csv` shows who the dense blocks leave out: the raw matrix has 536 models and 1289 tasks (12% observed), `mteb-dense56` keeps 28% of the models, 4% of the tasks and 10% of the observed cells, `mteb-sparse` 89%, 37% and 80%. A held-out model reveals the first k benchmarks of the selector's ordering for which it has scores and models with fewer than k scores are skipped for that k, so large-k points rest on fewer models and $k = 48$ on a small subset; targets are the observed cells of the fixed target benchmarks. Column-mean nMSE is 1.06; the full-information floors are 0.18 (factor posterior, 95% CI 0.15–0.20), 0.23 (kNN) and 0.43 (kernel ridge, ridge: these rules complete the matrix by mean imputation, so G against their own floor is lenient and the floor of the best rule is the honest reference). nMSE with conditional variance / random selection ($k = 2/4/8/16/32$): factor posterior $0.59/0.45/0.37/0.30/0.22$ and $0.65/0.52/0.42/0.33/0.26$; kernel ridge $0.44/0.35/0.34/0.29/0.22$ and $0.53/0.45/0.36/0.32/0.30$. Paired differences to ran-

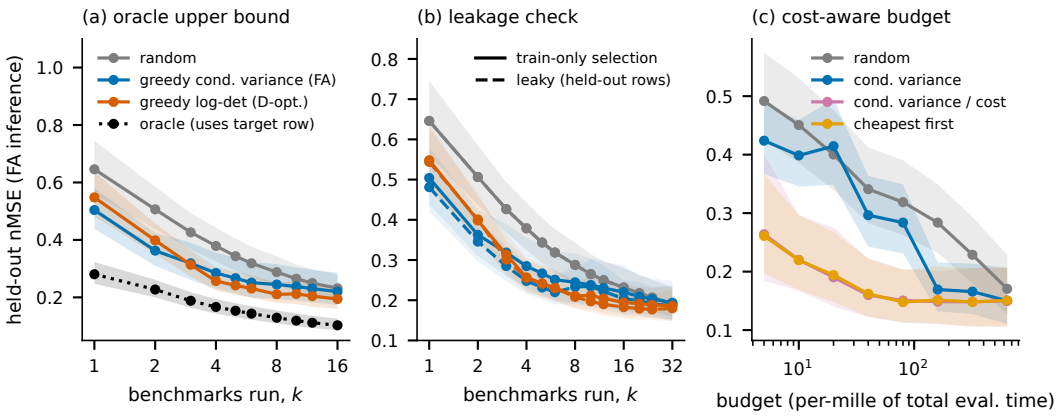


Figure 7: MTEB-56. (a) Oracle selection, which uses the held-out model's own target scores (optimistic upper bound, not a method). (b) Leakage check: the selection model fitted on training models only (solid, used everywhere) against also on the held-out rows (dashed). Panels (a, b) use factor-analysis inference, panel (c) kernel ridge; the text and Tables 6, 7 give kernel ridge. (c) Cost-aware selection: the budget is a share, in per mille, of the total evaluation time of the candidate tasks.

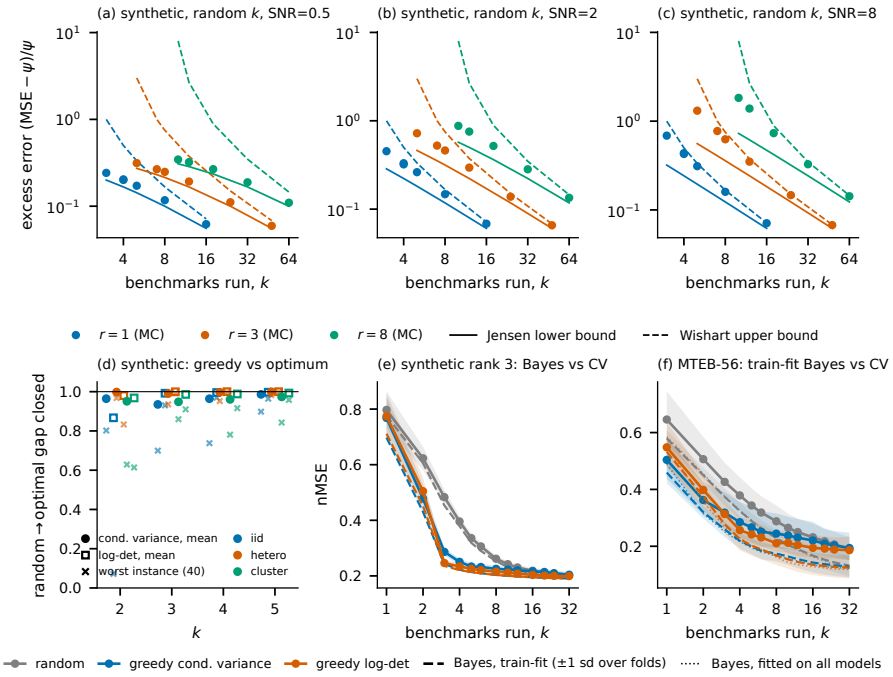


Figure 8: (a–c) Synthetic matrices with known truth: excess error over the noise floor for random k under Bayes-optimal inference (Monte Carlo) with the lower and upper bounds of Proposition 1. (d) Synthetic, $N = 16$: fraction of the random-to-optimal gap closed by the greedy rules (mean over 40 matrices, worst matrix). (e, f) Model-implied (dashed) and cross-validated (solid) nMSE for a synthetic rank-3 matrix and for MTEB-56; for MTEB-56 the dashed curve uses a covariance fitted on each fold's training models only (band: ± 1 sd of the model-implied error over folds), and the dotted curve the earlier in-sample fit, which is optimistic.

dom (model-level CI): conditional variance with the factor posterior $-0.047/-0.070/-0.083/-0.059/-0.024/+0.002$ at $k = 1/2/4/8/16/32$ (CI excludes 0 up to $k = 8$); log-det is worse than random for $k \leq 4$ ($+0.18$, $+0.17$, $+0.045$ at $k = 1, 2, 4$) and indistinguishable from $k = 8$; central is better at $k = 1, 2$ (-0.135, -0.047) and worse from $k = 4$ ($+0.039$ to $+0.137$). $k^\star(G \geq 0.8)$

Table 8: SYNTHETIC matrices only: nMSE of the column mean and of random, greedy conditional-variance (GV) and greedy log-det (D-opt) selection with factor-analysis inference (base: rank 3; noisy; clustered; heterogeneous; rank 8; $M = 30$ training models).

SYNTHETIC matrix	k	col. mean	random	GV	D-opt
syn. M30	2	1.15	0.75 ±0.14	0.50 ±0.13	0.50 ±0.13
	4	1.15	0.52 ±0.08	0.36 ±0.05	0.34 ±0.04
	8	1.15	0.38 ±0.04	0.35 ±0.05	0.32 ±0.04
syn. base	2	1.03	0.62 ±0.04	0.48 ±0.05	0.50 ±0.05
	4	1.03	0.40 ±0.02	0.25 ±0.01	0.23 ±0.01
	8	1.03	0.26 ±0.01	0.23 ±0.01	0.22 ±0.01
syn. clust	2	1.03	0.70 ±0.06	0.44 ±0.02	0.46 ±0.03
	4	1.03	0.52 ±0.03	0.41 ±0.02	0.42 ±0.02
	8	1.03	0.42 ±0.02	0.40 ±0.02	0.38 ±0.02
syn. hetero	2	1.03	0.78 ±0.07	0.58 ±0.05	0.66 ±0.08
	4	1.03	0.60 ±0.04	0.38 ±0.02	0.37 ±0.02
	8	1.03	0.45 ±0.02	0.38 ±0.02	0.37 ±0.02
syn. miss50	2	1.07	0.65 ±0.06	0.64 ±0.08	0.54 ±0.07
	4	1.07	0.42 ±0.03	0.49 ±0.06	0.30 ±0.02
	8	1.07	0.29 ±0.02	0.33 ±0.03	0.26 ±0.02
syn. noisy	2	1.03	0.79 ±0.05	0.68 ±0.05	0.66 ±0.05
	4	1.03	0.66 ±0.03	0.51 ±0.02	0.51 ±0.03
	8	1.03	0.53 ±0.02	0.49 ±0.02	0.47 ±0.02
syn. r8	2	1.03	0.85 ±0.07	0.82 ±0.07	0.83 ±0.08
	4	1.03	0.68 ±0.05	0.63 ±0.06	0.58 ±0.05
	8	1.03	0.42 ±0.02	0.28 ±0.01	0.27 ±0.02

Table 9: $k^\star(g)$ with the CI taken from the bootstrap over models / over strict families (kernel ridge and factor posterior; >32: not reached on the grid), and the range of the point $k^\star$ over the repetitions (fold-assignment draws). "central": most-central selection.

			$k^\star$ (CI rule): model / family				point $k^\star$ over repetitions			
dataset	rule	g	rand.	GV	D-opt	central	rand.	GV	D-opt	central
mteb-dense56	krr	0.8	5/5	2/3	3/3	4/5	4	2	2–3	2–3
	krr	0.9	12/12	8/8	4/6	10/10	8–10	3–8	3–4	5–8
	fa	0.8	8/8	4/4	3/4	6/6	5–6	2–3	3	4–5
	fa	0.9	16/16	20/10	5/6	10/12	10–12	4–12	4–6	8
mteb-dense	krr	0.8	3/3	1/1	2/2	2/2	3	1	2	1
	krr	0.9	6/6	5/5	3/4	12/16	4	2	2	2
	fa	0.8	10/10	6/8	8/8	24/32	8	3	4–5	12–20
	fa	0.9	32/32	20/16	20/16	>32/>32	20–24	12	10–12	32
mteb-mid	krr	0.8	3/3	2/3	2/2	5/6	2–3	2	2	2–3
	krr	0.9	5/4	3/3	3/3	8/10	3–5	2–3	2–3	3–6
	fa	0.8	16/16	8/8	12/10	>32/>32	12–16	6	8–10	>32
	fa	0.9	>32/>32	24/32	32/32	>32/>32	32	16–24	20	>32
mteb-sparse	krr	0.8	3/3	2/2	4/4	4/4	1–5	1–3	2–4	1–24
	krr	0.9	5/5	3/3	4/4	>32/16	2–10	1–4	2–5	1
	fa	0.8	16/16	16/16	16/24	>32/>32	12–16	8–12	12	>32
	fa	0.9	>32/>32	>32/32	24/24	>32/>32	32	20–32	20–24	>32
helm-mmlu	krr	0.8	2/2	1/1	3/3	1/1	1–2	1	1–3	1
	krr	0.9	3/2	1/1	3/3	1/1	1–3	1–2	1–3	1
	fa	0.8	3/3	1/2	3/3	1/1	2–3	1	1–3	1
	fa	0.9	5/5	3/3	4/4	16/16	3–5	2–3	3–4	1–4

with the factor posterior (`kstar_table.csv`): 16 for conditional variance and random, 24 for log-det; at $g = 0.9$: 32, 48, 24. With kernel ridge the criterion is reached at 2 (conditional variance), 4 (log-det), 3 (random) against its own floor and at 12, 12, 10 against the best floor. In the cluster-bootstrap run (1000 resamples, `cluster_kstar_B_mteb_sparse.csv`) the factor-posterior $k^\star$ is 16 for conditional variance, log-det and random under model resampling and 16, 24, 16 under family resampling, with point values 12, 12, 16 (the 24 for log-det in `kstar_table.csv` against 16 here shows the lower-bound rule grazing the criterion). The train-only Bayes curve is *above* the cross-validated error at $k \leq 4$ (conditional variance: 0.67, 0.53 against 0.59, 0.45) and below it from $k = 16$ (0.17 against 0.30), so the model-implied curve does not bound the error on this matrix. Two repetitions only.

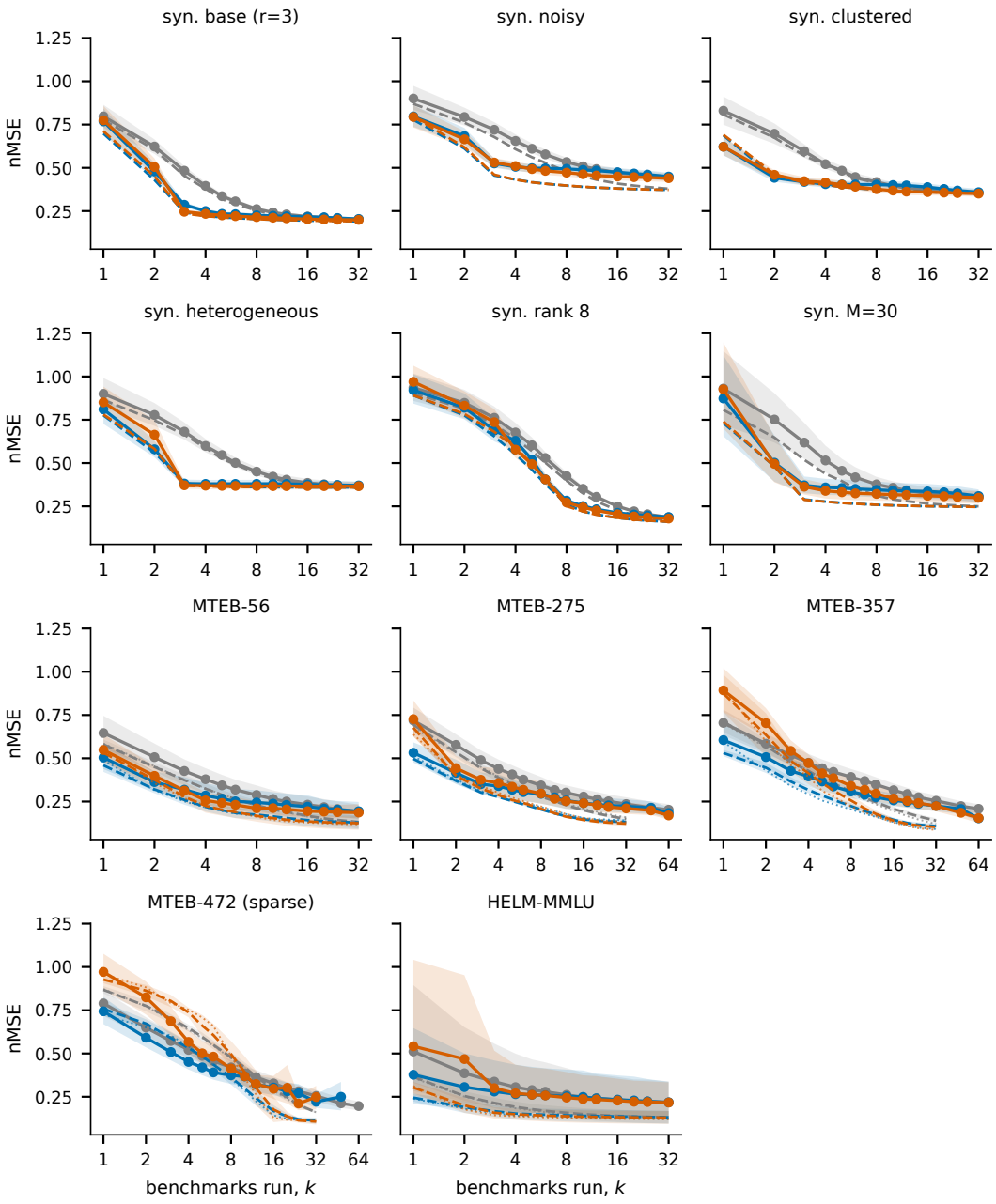


Figure 9: Panels (e, f) of Fig. 8 for every matrix with a model-implied curve (synthetic matrices are marked "syn."); real matrices: dashed, covariance fitted on training folds (± 1 sd over folds); dotted, in-sample fit.

C Reproduction notes

The repository root `README.md` gives the complete notes; this section lists what a reader needs to check the paper.

Data pins. MTEB results at commit `6dabbcc6...`; Open LLM Leaderboard `open-llm-leaderboard/contents` at `9c09a7ca...`; HELM Lite and HELM MMLU v1.13.0; LiveBench `model_judgment` at `9704e5da...` (full SHAs in `README.md`). `code/fetch_data.sh` fetches them (≈ 3.3 GB); the HELM Lite matrix with 91×28 entries is

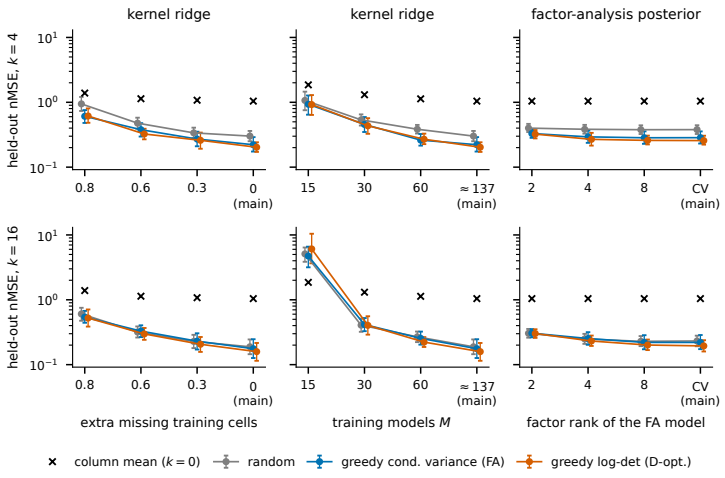


Figure 10: Sensitivity on MTEB-56 at $k = 4$ (top) and $k = 16$ (bottom): extra missing-at-random training cells, number of training models M, and fixed factor rank. Crosses: column-mean error under the same condition; bars: 95% bootstrap CI.

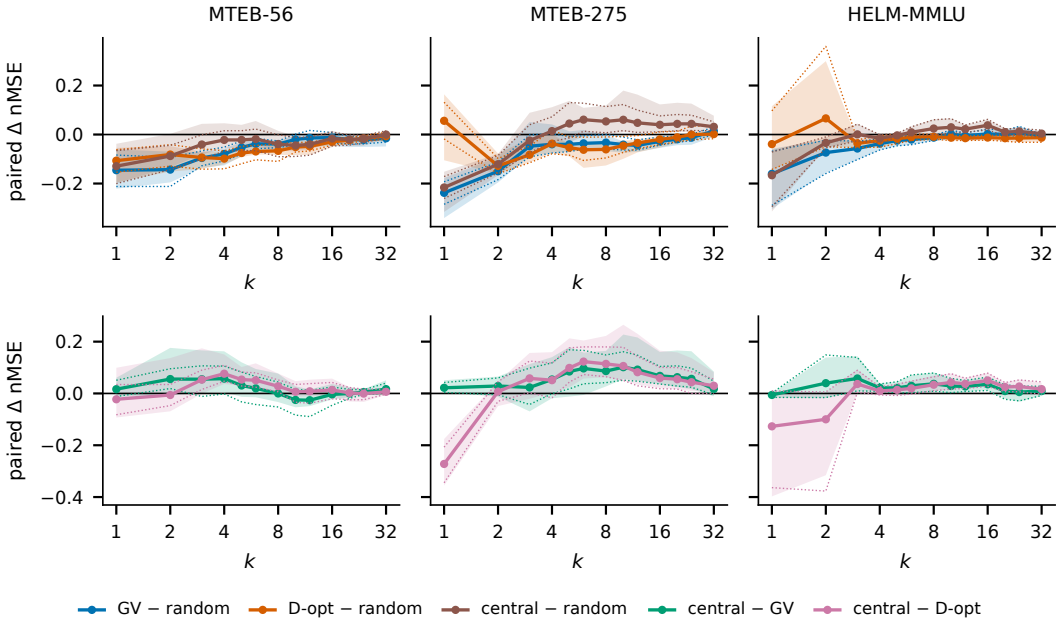


Figure 11: Paired differences in nMSE between selectors (kernel ridge, random folds; columns: MTEB-56, MTEB-275, HELM-MMLU; below 0 favours the first-named selector; top: conditional variance, log-det and central against random, bottom: central against conditional variance and log-det). Shaded: 95% CI from a bootstrap over strict families; dotted: bootstrap over models (same 1000 resamples).

built from the pinned HELM Lite v1.13.0 export by `code/build_helm_ext.py`. The pinned Open LLM Leaderboard export has six aggregate benchmarks (no per-subtask scores) and the pinned LiveBench export seven task-level scores; per-subtask leaderboard details are not part of the pinned material and were not fetched, so no larger generative-LLM matrix than HELM Lite (91×28) and HELM-MMLU (79×57) could be built.

Software and hardware. Python 3.11.2, numpy 2.4.6, scipy 1.17.1, pandas 3.0.6, pyarrow 26.0.0 (`code/requirements.txt`); CPU only. Wall-clock times on a shared host: MTEB-56 sweeps

Table 10: $k^\star(G \geq 0.8)$ by selector (kernel ridge and factor posterior). Random folds: CI rule with the bootstrap over models, organisations or strict families (1000 resamples), each with the $[2.5, 97.5]\%$ range of the point $k^\star$ over the resamples. Right: the same criterion (bootstrap over models) under leave-organisation-out, strict-family leave-out and the temporal split (single fold, MTEB only). "–": selector not run on that split. Under shift the floor and the column mean are far higher (temporal MTEB-56: 1.4–1.5 and 4.6), so G is relative to a poor floor.

			random folds: $k^\star$ [95% interval of point $k^\star$]			shift splits		
matrix	rule	selector	model CI	org. CI	family CI	org. out	family out	temporal
MTEB-56	krr	random	5 [3–5]	5 [3–5]	5 [3–5]	6	6	6
		GV	2 [2–2]	3 [1–3]	3 [2–3]	8	2	4
		D-opt	3 [2–3]	3 [2–3]	3 [2–3]	3	8	2
		central	4 [2–4]	5 [2–5]	5 [2–5]	–	10	–
	fa	random	8 [4–8]	8 [4–8]	8 [4–8]	8	8	5
		GV	4 [2–4]	5 [2–5]	4 [2–4]	16	5	4
		D-opt	3 [3–3]	4 [2–4]	4 [2–4]	4	6	2
		central	6 [3–6]	6 [3–6]	6 [3–6]	–	10	–
MTEB-275	krr	random	3 [3–3]	4 [2–4]	3 [2–3]	4	3	6
		GV	1 [1–1]	2 [1–2]	1 [1–1]	5	1	4
		D-opt	2 [2–2]	2 [2–2]	2 [2–2]	–	2	–
		central	2 [1–2]	2 [1–2]	2 [1–2]	–	1	–
	fa	random	10 [6–10]	10 [6–10]	10 [6–10]	10	8	16
		GV	6 [3–6]	6 [3–6]	8 [2–8]	6	5	8
		D-opt	8 [3–8]	8 [3–8]	8 [3–8]	–	3	–
		central	24 [10–24]	24 [10–24]	32 [10–32]	–	20	–
MTEB-472	krr	random	3 [2–3]	4 [2–4]	3 [1–3]	–	–	–
		GV	2 [1–2]	2 [1–2]	2 [1–2]	–	–	–
		D-opt	4 [2–4]	4 [2–4]	4 [1–4]	–	–	–
		central	4 [1–4]	4 [1–4]	4 [1–4]	–	–	–
	fa	random	16 [12–16]	20 [12–20]	16 [10–16]	–	–	–
		GV	16 [8–16]	16 [8–16]	16 [8–16]	–	–	–
		D-opt	16 [10–16]	16 [10–16]	24 [9–16]	–	–	–
		central	>32 [32–32]	>32 [32–32]	>32 [24–32]	–	–	–
HELM-MMLU	krr	random	2 [1–2]	2 [1–2]	2 [1–2]	–	–	–
		GV	1 [1–1]	1 [1–1]	1 [1–1]	–	–	–
		D-opt	3 [1–3]	3 [1–3]	3 [1–3]	–	–	–
		central	1 [1–1]	1 [1–1]	1 [1–1]	–	–	–
	fa	random	3 [2–3]	3 [2–3]	3 [2–3]	–	–	–
		GV	1 [1–1]	2 [1–2]	2 [1–2]	–	–	–
		D-opt	3 [1–3]	3 [1–3]	3 [1–3]	–	–	–
		central	1 [1–1]	2 [1–2]	1 [1–1]	–	–	–

20–55 min, MTEB-275/357 sweeps 1.5–3 h, small-N matrices minutes; CPU-hours were not metered.

Seeds. `--seed 0`; repetition i uses `default_rng(1000*seed+`i`)` and factor-analysis initialisation seed i; 3 repetitions of 10-fold CV on MTEB-56, 2 on MTEB-275/357, 5 on HELM-MMLU; 10 random-selector draws per fold (5 for the Open LLM Leaderboard). Bootstrap: 500 resamples.

Script to table or figure. `code/run_cv.py` (one run per command in `code/launch_sweeps.sh`) → `results/raw` → `code/analyze.py` → `results/summary`, `results/theory`; `code/figtab/make_all.sh` writes every figure and table from those CSVs (map in `figures/PROVENANCE.md`). Every run has a launch line: `code/launch_primary.sh` (`B_*`, `A_*`, `F_*`), `launch_sweeps.sh`, `launch_syn.sh` and `launch_review.sh` (cluster bootstrap, extra selectors and rules, shift rules, detector); `LOCAL=1 bash code/launch_primary.sh` runs them sequentially in plain Python and `bash code/check.sh` regenerates one small run in about 12 s and compares it with `results/check/expected.csv`. The raw per-model tables (450 MB compressed) are the release `raw-results-1` of the experiment repository.

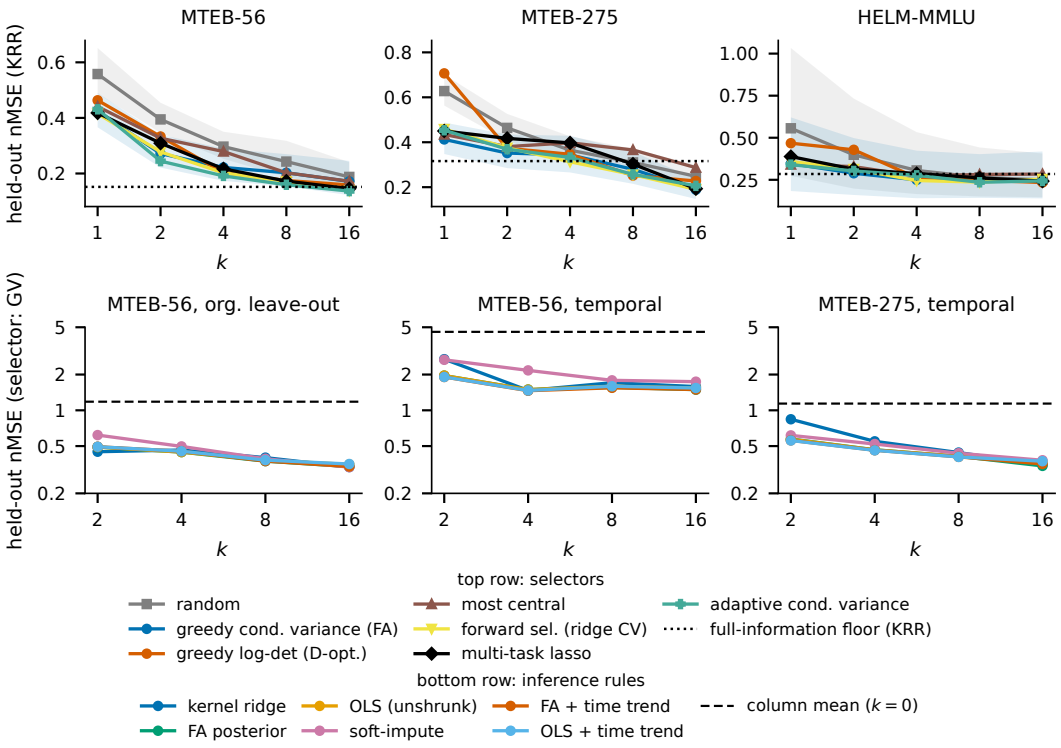


Figure 12: Top: kernel-ridge nMSE against k for the base selectors and three further deployable selectors (bands: 95% bootstrap CI over models for random and conditional variance only; dotted: full-information floor). Bottom: inference rules under organisation leave-out and temporal splits (conditional-variance set, log axis; dashed: column mean): kernel ridge, factor posterior, unshrunk OLS, soft-impute and the factor/OLS rule with a linear release-date trend.

Not finished. Every primary run listed above finished. Per-run wall-clock was not logged. The raw per-model tables are a release asset (`raw-results-1`, sparse run included), not versioned in the repository.

Forbidden-content log. The task excludes the published paper on this question and anything citing it. No search made for this study (context review, reference verification) returned such a work; none was read or used. A reviewer pointed to two works on low-rank factors and adaptive calibration for LLM evaluation; because they may fall under the exclusion they were not read or cited.

D VOLUNTARY SELF-AUDIT (NOT AN ICLR FORM)

ICLR 2027 has no mandatory checklist; we answer the following items for checkability.

1. *Claims match the evidence and its scope.* Yes: Sections 5, 6; scope is embedding-model matrices (MTEB) plus five small generative-LLM matrices and one sparse MTEB matrix, random-fold and shift splits.
2. *Limitations and failure cases.* Yes: Section 6 (temporal extrapolation, small M, sparsity, small-N leaderboards, cost-aware instability).
3. *Theory: assumptions, proofs, check.* Yes: Appendix A; checked on synthetic and real matrices (Fig. 8).
4. *Setting specified.* Yes: Section 4, Appendix C.
5. *Uncertainty.* Yes: 95% bootstrap over held-out models for every curve and paired difference; shift splits have few held-out models and are marked.
6. *Compute.* Yes: CPU only, times in Appendix C.

Table 11: Selectors with kernel-ridge inference, random folds, same folds and repetitions as the base runs: nMSE $\pm$ half-width of the 95% bootstrap CI over models.

	nMSE (KRR), number of benchmarks run k				
selector	1	2	4	8	16
MTEB-56					
random	0.56 $\pm$0.08	0.39 $\pm$0.05	0.30 $\pm$0.05	0.24 $\pm$0.07	0.19 $\pm$0.05
central	0.44 $\pm$0.07	0.33 $\pm$0.05	0.28 $\pm$0.05	0.20 $\pm$0.05	0.17 $\pm$0.04
GV	0.42 $\pm$0.06	0.27 $\pm$0.06	0.22 $\pm$0.06	0.20 $\pm$0.06	0.17 $\pm$0.06
D-opt	0.46 $\pm$0.07	0.33 $\pm$0.05	0.20 $\pm$0.04	0.17 $\pm$0.04	0.16 $\pm$0.05
adaptive GV	0.43 $\pm$0.06	0.24 $\pm$0.04	0.19 $\pm$0.03	0.16 $\pm$0.03	0.14 $\pm$0.03
fwd. ridge	0.42 $\pm$0.06	0.28 $\pm$0.05	0.20 $\pm$0.03	0.17 $\pm$0.04	0.14 $\pm$0.04
multi-task lasso	0.42 $\pm$0.06	0.31 $\pm$0.06	0.22 $\pm$0.04	0.17 $\pm$0.04	0.15 $\pm$0.04
MTEB-275					
random	0.63 $\pm$0.07	0.46 $\pm$0.06	0.36 $\pm$0.06	0.31 $\pm$0.06	0.25 $\pm$0.05
central	0.43 $\pm$0.08	0.38 $\pm$0.08	0.40 $\pm$0.09	0.37 $\pm$0.09	0.29 $\pm$0.07
GV	0.41 $\pm$0.07	0.35 $\pm$0.07	0.34 $\pm$0.08	0.28 $\pm$0.07	0.19 $\pm$0.04
D-opt	0.71 $\pm$0.11	0.37 $\pm$0.07	0.35 $\pm$0.09	0.25 $\pm$0.06	0.23 $\pm$0.06
adaptive GV	0.45 $\pm$0.08	0.37 $\pm$0.08	0.33 $\pm$0.09	0.26 $\pm$0.06	0.21 $\pm$0.04
fwd. ridge	0.46 $\pm$0.08	0.38 $\pm$0.08	0.31 $\pm$0.07	0.26 $\pm$0.06	0.19 $\pm$0.04
multi-task lasso	0.45 $\pm$0.07	0.42 $\pm$0.08	0.40 $\pm$0.09	0.30 $\pm$0.07	0.19 $\pm$0.04
HELM-MMLU					
random	0.56 $\pm$0.38	0.40 $\pm$0.27	0.31 $\pm$0.18	0.26 $\pm$0.15	0.25 $\pm$0.13
central	0.34 $\pm$0.22	0.33 $\pm$0.23	0.27 $\pm$0.15	0.28 $\pm$0.16	0.28 $\pm$0.14
GV	0.35 $\pm$0.22	0.29 $\pm$0.17	0.25 $\pm$0.14	0.24 $\pm$0.13	0.25 $\pm$0.14
D-opt	0.47 $\pm$0.40	0.43 $\pm$0.37	0.26 $\pm$0.14	0.25 $\pm$0.14	0.23 $\pm$0.12
adaptive GV	0.34 $\pm$0.22	0.30 $\pm$0.19	0.28 $\pm$0.16	0.24 $\pm$0.12	0.24 $\pm$0.13
fwd. ridge	0.36 $\pm$0.23	0.31 $\pm$0.19	0.25 $\pm$0.13	0.24 $\pm$0.13	0.25 $\pm$0.14
multi-task lasso	0.39 $\pm$0.28	0.31 $\pm$0.20	0.29 $\pm$0.17	0.26 $\pm$0.15	0.24 $\pm$0.14

Table 12: Differences in nMSE to random (left) and to conditional variance (right), CI from the bootstrap over strict families; $*$: CI excludes 0. Against conditional variance the family-cluster CI excludes 0 in 21 of 75 (selector $\times k$) cells: in 15 conditional variance is better (central in 7, log-det at $k = 1$ on MTEB-275 by 0.29, the rest by ≤ 0.09), in 6 the challenger is better by ≤ 0.04.

	Δ nMSE vs random					Δ nMSE vs GV				
selector	1	2	4	8	16	1	2	4	8	16
MTEB-56										
central	-0.12^*	-0.07	-0.02	-0.04	-0.02	0.02	0.06	0.06^*	-0.00	-0.01
GV	-0.13^*	-0.12^*	-0.07^*	-0.04^*	-0.01	–	–	–	–	–
D-opt	-0.09^*	-0.06^*	-0.09^*	-0.07^*	-0.03^*	0.04	0.06	-0.02	-0.03	-0.01
adaptive GV	-0.13^*	-0.15^*	-0.11^*	-0.08^*	-0.05^*	0.01	-0.03	-0.03	-0.04^*	-0.04^*
fwd. ridge	-0.14^*	-0.12^*	-0.09^*	-0.07^*	-0.05^*	-0.01	0.01	-0.02	-0.03	-0.04
multi-task lasso	-0.14^*	-0.09^*	-0.08^*	-0.07^*	-0.04^*	-0.01	0.04	-0.01	-0.03	-0.03^*
MTEB-275										
central	-0.19^*	-0.08^*	0.03	0.06	0.04^*	0.02^*	0.03	0.05	0.09^*	0.05^*
GV	-0.22^*	-0.11^*	-0.02	-0.03	-0.02^*	–	–	–	–	–
D-opt	0.08	-0.09^*	-0.02	-0.06^*	-0.02^*	0.29^*	0.02	0.00	-0.03	0.01
adaptive GV	-0.17^*	-0.09^*	-0.03	-0.05^*	-0.04^*	0.04^*	0.02	-0.01	-0.02	-0.02
fwd. ridge	-0.17^*	-0.09^*	-0.05^*	-0.05^*	-0.05^*	0.05^*	0.02	-0.03^*	-0.02	-0.02^*
multi-task lasso	-0.18^*	-0.05	0.03	-0.01	-0.03^*	0.04	0.07^*	0.05	0.02	-0.01
HELM-MMLU										
central	-0.22^*	-0.07^*	-0.03^*	0.02	0.03^*	-0.01	0.04	0.02^*	0.04^*	0.04^*
GV	-0.21^*	-0.11^*	-0.06^*	-0.02^*	-0.00	–	–	–	–	–
D-opt	-0.09^*	0.03	-0.04^*	-0.02^*	-0.02^*	0.12	0.14	0.01^*	0.00	-0.01
adaptive GV	-0.22^*	-0.10^*	-0.03^*	-0.03^*	-0.01^*	-0.01^*	0.01	0.03^*	-0.01	-0.01
fwd. ridge	-0.20^*	-0.08^*	-0.06^*	-0.03^*	0.00	0.01	0.03^*	-0.00	-0.01	0.01
multi-task lasso	-0.17^*	-0.09^*	-0.02^*	-0.00	-0.01	0.04	0.02	0.04^*	0.02	-0.01

7. *Authorship accountability (ICLR AI policy).* No: no human author exists and none reviewed the work; see the AI-use statement.

8. *Code and data.* Yes: scripts, versions, commands, map; licences: several source leaderboards declare none (ethics statement).

9. *Every number traces to a script.* Yes: `results/summary`, `results/theory`, `figures/PROVENANCE.md`.

Table 13: Inference rules under random folds with random and conditional-variance selection: kernel ridge, factor posterior with CV rank, fixed rank 1 and 2, unshrunk OLS and soft-impute; nMSE $\pm$ half-width of the 95% CI over models.

	random selection				greedy cond. variance			
rule	2	4	8	16	2	4	8	16
MTEB-56								
kernel ridge	0.39±0.05	0.30±0.05	0.24±0.07	0.19±0.05	0.27±0.06	0.22±0.06	0.20±0.06	0.17±0.06
FA (CV rank)	0.48±0.06	0.36±0.05	0.29±0.06	0.23±0.04	0.36±0.06	0.28±0.06	0.24±0.06	0.22±0.06
FA, $r = 1$	0.54±0.07	0.48±0.07	0.43±0.06	0.41±0.06	0.46±0.07	0.42±0.06	0.40±0.06	0.38±0.06
FA, $r = 2$	0.49±0.06	0.39±0.06	0.33±0.05	0.30±0.05	0.37±0.06	0.33±0.05	0.31±0.05	0.29±0.05
OLS	0.48±0.06	0.36±0.05	0.30±0.08	0.25±0.07	0.36±0.06	0.28±0.06	0.24±0.06	0.21±0.06
soft-impute	0.59±0.09	0.45±0.07	0.34±0.05	0.27±0.04	0.50±0.09	0.37±0.05	0.31±0.04	0.26±0.05
MTEB-275								
kernel ridge	0.46±0.06	0.36±0.06	0.31±0.06	0.25±0.05	0.35±0.07	0.34±0.08	0.28±0.07	0.19±0.04
FA (CV rank)	0.54±0.06	0.42±0.05	0.34±0.05	0.28±0.04	0.42±0.07	0.34±0.06	0.29±0.06	0.21±0.04
FA, $r = 1$	0.63±0.06	0.56±0.06	0.52±0.05	0.50±0.05	0.54±0.07	0.50±0.06	0.49±0.05	0.47±0.05
FA, $r = 2$	0.55±0.06	0.44±0.06	0.38±0.05	0.35±0.05	0.41±0.07	0.36±0.06	0.34±0.05	0.29±0.04
OLS	0.54±0.06	0.42±0.05	0.34±0.05	0.29±0.05	0.42±0.07	0.34±0.06	0.29±0.06	0.21±0.04
soft-impute	0.64±0.07	0.50±0.06	0.40±0.05	0.32±0.04	0.57±0.07	0.45±0.06	0.35±0.05	0.26±0.03
HELM-MMLU								
kernel ridge	0.40±0.27	0.31±0.18	0.26±0.15	0.25±0.13	0.29±0.17	0.25±0.14	0.24±0.13	0.25±0.14
FA (CV rank)	0.43±0.27	0.33±0.18	0.27±0.14	0.23±0.11	0.31±0.16	0.27±0.14	0.25±0.13	0.23±0.12
FA, $r = 1$	0.46±0.31	0.40±0.26	0.36±0.24	0.34±0.23	0.38±0.23	0.37±0.25	0.36±0.23	0.35±0.23
FA, $r = 2$	0.43±0.27	0.32±0.18	0.27±0.15	0.24±0.12	0.30±0.16	0.28±0.16	0.27±0.15	0.25±0.14
OLS	0.43±0.27	0.33±0.18	0.28±0.15	0.28±0.14	0.30±0.16	0.27±0.15	0.26±0.14	0.27±0.16
soft-impute	0.55±0.35	0.41±0.26	0.33±0.21	0.27±0.16	0.52±0.39	0.37±0.22	0.30±0.17	0.26±0.15

Table 14: Out-of-range detector (selector: conditional variance; flag: squared Mahalanobis distance of the k revealed scores above the 95% χ^2 quantile of the shrunk training covariance). AUC: of the distance for the worst tercile of the model's own nMSE (kernel ridge); mean nMSE of flagged and unflagged models with 95% bootstrap CI over models (n/a: fewer than 3 flagged).

matrix, split	k	flagged	AUC	nMSE flagged	nMSE unflagged
MTEB-56, random folds	4	12/152	0.66	0.71 [0.23, 1.43]	0.18 [0.15, 0.21]
	8	11/152	0.82	0.83 [0.32, 1.60]	0.15 [0.13, 0.18]
	16	10/148	0.86	0.94 [0.34, 1.77]	0.12 [0.10, 0.14]
MTEB-56, org. leave-out	4	14/152	0.79	2.25 [0.97, 4.13]	0.28 [0.23, 0.34]
	8	12/152	0.80	2.44 [1.12, 4.14]	0.22 [0.19, 0.26]
	16	22/150	0.88	1.31 [0.79, 2.14]	0.17 [0.15, 0.20]
MTEB-56, family leave-out	4	7/152	0.73	2.01 [0.63, 4.88]	0.30 [0.26, 0.34]
	8	15/152	0.84	1.19 [0.55, 2.27]	0.23 [0.20, 0.25]
	16	21/148	0.84	0.93 [0.47, 1.83]	0.19 [0.16, 0.21]
MTEB-56, temporal	4	14/31	0.93	2.41 [1.66, 3.18]	0.73 [0.30, 1.46]
	8	12/31	0.96	3.62 [2.81, 4.32]	0.56 [0.37, 0.78]
	16	14/29	0.89	2.52 [1.65, 3.32]	0.75 [0.30, 1.33]
MTEB-275, random folds	4	1/133	0.72	2.13 [n/a]	0.33 [0.26, 0.41]
	8	7/133	0.85	1.48 [0.69, 2.21]	0.21 [0.18, 0.26]
	16	5/112	0.84	0.67 [0.16, 1.32]	0.17 [0.13, 0.20]
MTEB-275, org. leave-out	4	11/133	0.74	2.01 [1.39, 2.76]	0.32 [0.28, 0.37]
	8	12/133	0.83	1.36 [0.97, 1.95]	0.27 [0.23, 0.32]
	16	13/113	0.90	0.70 [0.58, 0.80]	0.20 [0.17, 0.23]
MTEB-275, temporal	4	1/27	0.70	0.58 [n/a]	0.55 [0.43, 0.71]
	8	2/27	0.68	1.33 [n/a]	0.37 [0.29, 0.45]
	16	3/15	0.96	0.65 [0.35, 0.97]	0.27 [0.18, 0.38]

10. *Citations.* Yes: all entries were checked against public metadata; forbidden-content log in Appendix C.
11. *Failed and unfinished runs reported.* Yes: Section 5.1, Appendix C.

Table 15: Temporal drift on MTEB-56 and MTEB-275 (dated models only). (a) Single temporal fold (train: earliest $4/5$, test: newest $\approx 1/5$; conditional-variance selection): nMSE $\pm$ half-width of the 95% CI over models for six inference rules, including a linear release-date trend added to the factor and OLS rules; the column mean ($k = 0$) is in the last row of the block. (b) Rolling origin (cutoffs at $55, 65, 75, 85\%$ of the dated models, the next 15% as test, pooled over the four origins): random against conditional-variance selection, with the column mean and the column mean plus trend at $k = 0$.

	MTEB-56				MTEB-275			
rule	2	4	8	16	2	4	8	16
(a) single temporal fold, selector GV, test = newest ≈20% of dated models								
FA	1.96±0.71	1.50±0.54	1.56±0.53	1.49±0.55	0.57±0.11	0.47±0.08	0.41±0.07	0.34±0.10
KRR	2.69±0.92	1.47±0.55	1.71±0.58	1.58±0.56	0.84±0.25	0.55±0.14	0.44±0.12	0.35±0.12
OLS (unshrunk)	1.96±0.71	1.49±0.53	1.60±0.53	1.54±0.56	0.57±0.11	0.46±0.08	0.41±0.07	0.37±0.13
soft-impute	2.66±0.96	2.17±0.78	1.79±0.59	1.74±0.62	0.62±0.11	0.52±0.08	0.44±0.07	0.38±0.09
FA + trend	1.91±0.68	1.46±0.53	1.55±0.53	1.50±0.56	0.56±0.11	0.46±0.07	0.41±0.07	0.35±0.10
OLS + trend	1.91±0.68	1.47±0.53	1.59±0.53	1.54±0.56	0.56±0.11	0.46±0.07	0.41±0.07	0.37±0.12
avg.-score regression	2.13±0.73	2.16±0.72	1.81±0.56	1.98±0.64	0.58±0.11	0.54±0.09	0.51±0.08	0.55±0.14
column mean ($k = 0$)		4.57				1.14		
(b) rolling origin (cutoffs 55/65/75/85%, next 15% as test), pooled								
FA, random	1.25±0.34	1.04±0.28	0.84±0.25	0.74±0.23	0.73±0.10	0.59±0.08	0.49±0.08	0.43±0.07
KRR, random	1.18±0.28	0.99±0.24	0.88±0.23	0.85±0.23	0.69±0.10	0.58±0.10	0.48±0.09	0.42±0.08
soft-impute, random	1.38±0.38	1.12±0.32	0.92±0.25	0.78±0.21	0.83±0.11	0.65±0.09	0.53±0.07	0.44±0.06
FA + trend, random	1.29±0.32	1.08±0.28	0.86±0.25	0.76±0.25	0.79±0.11	0.62±0.09	0.50±0.08	0.44±0.08
FA, GV	0.96±0.31	0.91±0.42	0.87±0.44	0.74±0.43	0.58±0.09	0.48±0.08	0.45±0.10	0.39±0.11
KRR, GV	1.15±0.28	1.03±0.41	0.94±0.39	0.93±0.42	0.56±0.12	0.51±0.11	0.46±0.12	0.38±0.11
soft-impute, GV	1.24±0.32	1.02±0.34	0.86±0.27	0.71±0.23	0.70±0.10	0.55±0.08	0.49±0.07	0.40±0.06
FA + trend, GV	0.98±0.29	0.93±0.49	0.90±0.46	0.78±0.47	0.59±0.09	0.49±0.09	0.47±0.11	0.40±0.11
column mean ($k = 0$)		2.18				1.28		
column mean + trend ($k = 0$)		2.03				1.37		